2023 - 2024

FUTURE
UNIVERSITY IN EGYPT
FACULTY OF
COMPUTERS AND INFORMATION TECHNOLOGY

2024-2023

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Introduction

The Faculty of Computers and Information Technology at Future University was established in 2006. Since its establishment, the faculty has been working to achieve excellence in the fields of computing and informatics and to keep pace with the continuous and rapid technological development in the job market.

Accordingly, stemming from its national responsibility and in pursuit of fulfilling its mission, the faculty is keen on preparing generations equipped with the foundations of modern knowledge and fully ready to keep up with rapid technological advancements. It also aims to develop students' ability to achieve competitive quality by performing their duties efficiently and sincerely. The curricula and courses offered are continuously evaluated, and the latest educational systems are adopted to allow greater student participation and to open up wider opportunities for students to choose courses according to their abilities, potentials, and interests.

Within this framework, this regulation has been developed in accordance with the latest developments in the labor market and the advancements in the field of computing and informatics.

Chapter One: General Regulations

Vision and Mission

University Vision

Future University aspires to achieve leadership locally and regionally and to become a pioneering model in education, scientific research, and community service at both the national and international levels.

University Mission

Future University is committed to providing world-class quality educational services to prepare graduates with scientific and innovative competencies that keep pace with labor market changes. It also aims to provide a motivating environment for the advancement of scientific research and contribute to achieving sustainable development goals and technological progress, while instilling professional values and ethics.

Collage Vision

Striving to achieve a distinguished position in the field of computing and informatics locally, regionally, and internationally.

Collage Mission

The Faculty of Computers and Information Technology at Future University in Egypt (FUE) is committed to providing a distinguished educational environment to prepare qualified cadres in various fields of technology and software, capable of competing and keeping pace with the continuous development in the local, regional, and international labor markets. It also aims to produce outstanding scientific research that contributes to the generation of knowledge and advanced technology, and to participate effectively in achieving sustainable development goals while adhering to core values and professional ethics.

Strategic Goals

- Develop existing programs and introduce new ones that meet labor market requirements.
- Provide effective and distinguished education, evaluation, and service systems that support students and graduates in acquiring professional competencies, skills, and the capacity for lifelong learning.
- Ensure the sustainability of local accreditation for all existing programs and seek local and international accreditation for newly established programs.
- Guarantee the adequacy and efficiency of the faculty's research output while supporting excellence and innovation in research activities.
- Increase support for scientific research to contribute to achieving sustainable development goals in accordance with research ethics and intellectual property rights.
- Achieve distinguished partnerships with local, regional, and international institutions to develop various faculty activities.
- Provide effective community services aligned with sustainable development goals.
- Establish continuing education programs that attract graduates and the external community, keeping pace with technological developments.
- Ensure the adequacy and efficiency of physical resources to meet local and international benchmarks.
- Develop physical resources and technological infrastructure, and contribute to developing software and systems for managing operations within the university.
- Enhance quality assurance and the performance of human resources.

General Skills of the Faculty Programs

Upon successful completion of any program offered by the college, the graduate will acquire the following general skills and be able to:

First: Knowledge and Understanding

- Comprehend the fundamental concepts and theories related to computing and information systems.
- Describe the modeling and simulation of computer systems.
- Identify the methodologies, practices, and tools used in the stages of computer software systems development.
- Discuss specifications and strategic planning for proposing problem solutions.
- Determine the usage standards and future development of computer systems.
- Recognize the latest technologies used to support computer processing and communications.
- Be familiar with outline testing techniques and methods for computer-based systems.
- Discuss management principles relevant to computing and information systems.
- Recognize the technical, ethical, and moral issues involved in the computing and information industry.
- Identify trends in computing and information research.
- Explain the functional requirements and constraints in the development of computer-based systems.

Second: Mental Skills

- Analyze and design solutions for computing problems, taking into account constraints and limitations.
- Implement computing and information solutions in academic disciplines.
- Define measurement standards for the deployment and evolution of computer systems.
- Prepare proposals and presentations for computing and information systems.
- Test and evaluate the functionality of computer and information systems.
- Critique a system considering costs, various quality attributes, and environmental impact.
- Be aware of the professional, ethical, legal, and moral issues related to computing and information.
- Critique a research paper in a specific field.

Third: Practical and Professional Skills

- Operate computer equipment in different physical environments.
- Use various computing techniques in project development and deployment.
- Install and maintain various support tools for construction and documentation.
- Perceive skills for information storage and retrieval in computing and information contexts.
- Acquire a range of essential research skills from various sources.
- Analyze, design, implement, test, maintain, and manage software systems.
- Assess risks and safety aspects related to computer and information systems.
- Process big data and extract conclusions.

Fourth: General and Transferable Skills

- Utilize a range of learning resources.
- Work effectively within a team, considering time and pressure management.
- Use information retrieval techniques.
- Apply communication skills and techniques in presentations and report writing for a variety of audiences using different methods and tools.
- Apply quantitative methods and skills in understanding and presenting cases.
- Effectively utilize general computing facilities.
- Appreciate the importance of continuous professional development and lifelong learning.

These skills are updated in accordance with labor market developments and forecasts, as well as the National Academic Reference Standards (NARS) issued by the National Authority for Quality Assurance of Education and Accreditation.

Subject (1): College Departments, Programs, and Academic Degrees

Collage Departments

1. **Department of Computer Science**
2. **Department of Information Systems**
3. **Department of Digital Media Technology**

Collage Programs

1. **Computer Science Program**
2. **Artificial Intelligence Program**
3. **Cyber Security Program**
4. **Information Systems Program**
5. **Data Science Program**
6. **Digital Media Technology Program**

Academic Degrees

1. **Bachelor of Computers and Information degree in Computer Science**
2. **Bachelor of Computers and Information degree in Artificial Intelligence**
3. **Bachelor of Computers and Information degree in Cyber Security**
4. **Bachelor of Computers and Information degree in Information Systems**
5. **Bachelor of Computers and Information degree in Data Science**
6. **Bachelor of Computers and Information degree in Digital Media Technology**

Courses Offered by the Scientific Departments

- Courses offered by the Department of Computer Science are those whose course codes begin with **CS**.
- Courses offered by the Department of Digital Media Technology are those whose course codes begin with **DM**.
- Courses offered by the Department of Information Systems are those whose course codes begin with **IS**.

Article (2): Study System at the College

- The study system is based on the credit hour system, where the credit hour is the unit of study measurement used to determine the weight of a course.
- The academic year consists of two regular semesters (Fall and Spring semesters) and a summer semester, which is optional for students. The summer semester is held according to the college's capabilities and for a financial fee determined by the university.
- The duration of regular semesters is 16-17 weeks, including the exam period.
- The summer semester is an intensive semester where the weekly study hours for a course are doubled. Its duration is 7-8 weeks, including the exam period.
- The maximum duration of study at the college is 8 academic years (16 regular semesters, Fall and Spring), noting that approved suspension semesters, approved by the College Council and the University Council, are not counted within the permitted semesters.
- Study at the college follows the credit hour system, where the total number of hours required to obtain a bachelor's degree is 132 credit hours. Final-year students prepare a project extending over two semesters. The specialized department councils determine its topic, and the College Council, based on the proposal of the specialized department councils, sets an additional period for the project after the written exam at the end of the academic year. It also forms committees for the practical and oral evaluation of the projects.
- Language of Instruction: English.
- Definition of the Credit Hour: The student receives a number of credit hours according to the number of contact hours per week. Contact hours are calculated as follows:
    - One lecture hour counts as one credit hour.
    - Two or three hours (tutorials or labs) count as one credit hour.

Article (3): Admission and Transfer Rules

Admission Requirements for the College

- The college accepts graduates of general secondary education specializing in Mathematics or Sciences, or their equivalent from other accredited certificates, in the same academic year for enrollment in the college, in accordance with the coordination rules of the Council of Private Universities.
- Science stream students must pass the course equivalent to the Math 2 course intended for Mathematics stream students in general secondary education.

Collage Transfer Requirements

Students may be transferred to the college from other university faculties or similar or dissimilar institutes, provided they have obtained the minimum total score required for admission to the college, in a manner that does not conflict with the rules and regulations set by the Supreme Council of Universities and the Council of Private and National Universities. In case a student's transfer from a corresponding faculty is accepted, course equivalency (mapping) shall be conducted to equate the courses they have studied with the courses offered by the department to which they are transferring, subject to the approval of the College Dean and University President. Transfer from one specialization to another within the college is permitted, provided the student meets the admission requirements for the new specialization. Course equivalency requires that the student's passing grade in the course is not less than **65%** .

Article (4): Graduation Requirements and Conditions for Awarding a Bachelor's Degree

Future University, upon the request of the Faculty of Computers and Information Technology, awards the Bachelor's degree in the college's programs (mentioned in Article 3 of the Bylaws) once the student fulfills the requirements for obtaining it, which are:

Obtaining the Bachelor's degree requires the student to successfully complete **132 credit hours** in no less than **6 regular semesters** (Fall and Spring), with a Cumulative Grade Point Average (CGPA) of no less than **2.00**.

The minimum period for graduation (obtaining a Bachelor's degree) is **3 academic years**, i.e., **6 regular semesters** (Fall and Spring).

Passing any requirements stipulated by the university's rules as prerequisites for graduation.

Article (5): Rules for Enrollment in Programs

The College Council determines the regulations governing the distribution of students among the offered programs based on a proposal from the scientific department councils.

Article (6): Attendance Rules

Student entry to the final exam requires achieving an attendance rate of no less than **75%** of the lectures and tutorials in each course. If a student's absence rate in a course exceeds **25%** without an acceptable excuse, the College Council may deny them entry to the final exam, and they will receive a grade of "zero" on the final exam score for that course. However, if the student submits an excuse acceptable to the College Council (within the period specified by the council), a grade of "Withdrawn" will be recorded for them in the course for which the excuse was submitted.

- The College Council, after consulting with the relevant department council and depending on the nature of the courses, may decide to teach one or more courses using a blended learning model. This entails the course being delivered **60% face-to-face** and **40% via distance education**, or any other specified ratio. This decision must be presented to the University's Education and Student Affairs Council for approval and subsequently submitted to the University Council for ratification.

Article (7): Academic Level Determination

Upon the student's enrollment in the college:

- A new student is registered at the **first level**.
- A student transferred from another college is registered at the level equivalent to the number of credit hours they have successfully completed at the college from which they are transferring, based on course equivalency and as specified in the following table.
- A student may move from one level to the next at the beginning of each semester, according to the credit hours they have passed, as detailed in the following table.

Number of Credit Hours Successfully Completed by the Student	Student Classification	Academic Level
From 0 to less than 26	Freshman	First
From 26 to less than 57	Sophomore	Second
From 57 to less than 93	Junior	Third
93 or more	Senior	Fourth

Article (8): Credit Hour Enrollment Limits per Semester

- **Minimum Enrollment:** The minimum number of credit hours for enrollment is **9 hours**. This minimum may be waived if the remaining hours required for the student's graduation that are offered are less than 9 hours.
- **Maximum Enrollment in Regular Semesters (Spring - Fall):**
    - **18 credit hours** for new students (non-transfer students from corresponding faculties) in the first semester of their enrollment at the college.
    - **21 credit hours** for students who have a CGPA at the beginning of the semester that is **greater than or equal to 3.00**, as well as for students graduating in that same semester.

- **18 credit hours** for students with a CGPA at the beginning of the semester **greater than or equal to 2.00 and less than 3.00**.
- **15 credit hours** for students with a CGPA at the beginning of the semester **greater than or equal to 1.00 and less than 2.00**.
- **12 credit hours** for students with a CGPA at the beginning of the semester **less than 1.00**.
- Enrollment in the **summer semester** is optional for the student. The maximum number of credit hours a student can enroll in is **9 credit hours**, or as determined by the Council of Private Universities.

Article (9): Registration, Drop/Add, and Withdrawal Periods

- The College Council determines the minimum number of students required for a course to be offered when courses are listed for registration.

- At the beginning of each semester, the student registers for the courses they select during the times specified by the college administration, before the start of regular classes, according to the following:

  - **Add & Drop**
    After the registration process is completed and classes begin, a student may drop one or more courses they had previously registered for, or add one or more courses to complete the permitted credit hours for that semester, with the approval of their academic advisor. This is allowed before the end of the **third week** from the start of classes.

  - **Withdrawal**
    A student may withdraw from registration in one or more courses with the approval of their academic advisor during the period specified by the University Council, while observing the minimum number of credit hours required. A grade of "Withdrawn" **(W)** is recorded for them, and they must retake the entire course, including both study and examination.

- A student is not permitted to register for a course for which they have not successfully completed the prerequisite.

Article (10): Incomplete Grade

If the student submits a force majeure excuse accepted by the College Council for not attending the final exam for any course, within a maximum of one week from the date of the final exam for the course, a grade of "Incomplete" **"I"** is recorded for him in this

course, provided that he has obtained at least 60% of the semester work grades and has not been denied entry to the final exams.

A student who receives an "Incomplete" grade is permitted to take only the final exam. The student's final grade is calculated based on the grade obtained in the final exam in addition to the previously obtained semester work grade. The student must take the final exam during the same academic year or the following academic year from when the course was marked "Incomplete"; otherwise, the grade will convert to "Withdrawn" **"W"** , and the student must retake the entire course, including study and examination, without counting the previously obtained semester work grade.

- If a student submits a compelling excuse accepted by the College Council for not attending the final exam of any course, within a maximum of one week from the date of the final exam for that course, but does not meet the condition of having obtained at least 60% of the semester work grades, a grade of "Withdrawn" **"W"** will be recorded for them in the course, and they must retake the entire course, including study and examination.

- **Article (11): Suspension of Enrollment and Apology for One or More Semesters**

- A student is considered to have interrupted their studies if they do not register for a semester or withdraw from it without an excuse or with an unacceptable excuse.
- A student may suspend their enrollment with an excuse acceptable to the college and university for a maximum of **4 consecutive semesters** or **6 separate semesters** (even if some are consecutive). The student will be dismissed from the college if they interrupt their studies for a longer period without an excuse accepted by the college and approved by the university.
- A student may request to suspend their enrollment at the college according to the terms and conditions set by the university.

Article (12): Academic Advising

- The Vice Dean for Education and Student Affairs, in coordination with the heads of the scientific departments, assigns an academic advisor to each student to assist and guide them academically throughout their studies. This guidance concerns the selection of compulsory and elective courses and their prerequisites, which the student can register for in any regular or summer semester.
- The academic advisor's opinion is advisory. The student is fully responsible for the courses they register for based on their own choices.

Article (13): Examination System, Grades, and Course Points

• The minimum passing grade for a course is **50%** of the total course score, provided that the student obtains at least **30%** of the final exam score.

• **Exam grades for each course are distributed as follows:**

- Student performance is evaluated continuously during the semester (through assessment of participation and midterm written exams) in addition to the final written exam at the end of the semester (regular and summer). **Semester work represents 60%** of the final course grade.
- The final written exam at the end of each semester represents **40%** of the final grade.
- The duration of the final exam for each course shall not be less than two hours for credit hour courses.

• **Table of Grades and Points for Credit Hour Courses**

Points	Grade	Percentage Score
4.00	A+	96% or more
3.70	A	92% - less than 96%
3.40	A-	88% - less than 92%
3.20	B+	84% - less than 88%
3.00	B	80% - less than 84%
2.80	B-	76% - less than 80%
2.60	C+	72% - less than 76%
2.40	C	68% - less than 72%
2.20	C-	64% - less than 68%
2.00	D+	60% - less than 64%
1.50	D	55% - less than 60%
1.00	D-	50% - less than 55%
0.00	F	Less than 50%
0.00	**Abs**	Absence from the final exam without an excuse acceptable to the College Council
- - -	**Con**	Continuing course (without points, and the credit hours are not included in the cumulative average until the course is completed, either by passing or failing). The student is permitted to register for the course if the prerequisite has an "Incomplete" grade.
	I	Incomplete Course
	W	Withdrawal from a Course
-	**T**	Grade given for courses transferred by the student from another university

- **Pass/Fail Courses**

Grade	Means
P	Pass
F	FAIL
W	Withdrawn
Abs	Absence from the final exam without an excuse acceptable to the College Council
I	Incomplete (if the course has semester work)

- **Overall Grade Table (Per Semester / Upon Graduation)**

Overall Grade	Cumulative GPA (CGPA)
Very Poor	Less than 1.00
Poor	1.00 – Less than 2.00
Acceptable	2.00 – Less than 2.50
Good	2.50 – Less than 3.00
Very Good	3.00 – Less than 3.50
Excellent	3.50 or more

Article (14): Semester Average and Cumulative Average

$$Cumulated\ GPA\ (CGPA) = \frac{\sum_{\forall\ courses} Grade\ Points * Number\ of\ Credit\ hours}{\sum_{\forall\ courses} Number\ of\ Credit\ hours}$$

- The student's semester average (GPA) is calculated based on the sum of points the student obtained in all courses they registered for in the semester, divided by the total credit hours for those courses. The student's cumulative average (CGPA) is calculated based on the sum of points the student obtained in all courses they have studied, divided by the total credit hours for those courses. The student's grades are recorded in their academic record each semester, along with their semester and cumulative averages.
- The average shall be calculated to the nearest two decimal places.

Article (15): Graduation Project

• Each scientific department proposes topics related to its specialization for which graduation projects can be prepared and announces them to students. The graduation project is calculated as **6 credit hours** over two consecutive regular semesters (Fall and Spring). The rules for registering for the graduation project are as follows:

- A student who has successfully completed **93 credit hours** of the total credit hours required for graduation is entitled to register for the project course. A student expected to graduate after studying a summer semester who has not yet completed 93 credit hours may register for the project course after approval from the scientific department and the college.
  • The minimum passing grade for the project course is **50%** of the total course score, with the project discussion grades comprising **40%** .

Article (16): Academic Probation

• A student is placed on the academic probation list if their cumulative average falls below **2.00** (out of 4.00), except for the first semester. The student and their guardian are notified in writing of the probation status and its consequences. The academic probation is lifted if the student succeeds in raising their cumulative average to at least (2.00) in the semester following the probation.

Article (17): Course Repetition

• • Repeating a course previously failed in:

- If a student fails a course, they must retake it and be examined in it again. If they pass the course after retaking it, the actual grade they obtain is calculated for them, not exceeding **83** (the highest grade in **B**).
- Their cumulative average is calculated on this basis, counting the course hours only once.
- All instances of repetition and the grade (or evaluation) obtained by the student in each repetition appear on the student's detailed transcript.

• Repeating a course previously passed with points less than 2.00 to raise the cumulative GPA to avoid dismissal:

- If a student has a cumulative GPA (CGPA) at the beginning of the semester of less than 2.00 (student under academic probation), they must raise their average.
- The student mentioned in the previous point (placed under academic probation) is entitled to repeat any course they have previously passed (with points less than 2.00) to raise their cumulative GPA to avoid dismissal. In this case, they must retake the course and be examined in it again. They will receive the highest grade obtained from all repetition attempts, not exceeding **83** (the highest grade in **B**).

- There is no maximum limit for these courses; rather, the student (placed under academic probation) may repeat any number of courses they have previously passed (with points less than 2.00) in order to raise their cumulative GPA (CGPA) to at least 2.00.
- The student's cumulative average is calculated on this basis, counting the course hours only once.
- All instances of repetition and the grade (or evaluation) obtained by the student in each repetition appear on the student's detailed transcript.

• Repeating a course previously passed with points greater than or equal to 2.00 to raise the cumulative GPA for improvement:

- A student with a cumulative GPA greater than or equal to 2.00 is entitled to improvement to raise their cumulative GPA, but only if they have not yet completed the requirements for graduation.
- If a student wishes to repeat a course they have previously passed (with points greater than or equal to 2.00) to raise their cumulative GPA, they must retake the course and be examined in it again. In this case, they will receive the highest grade obtained from all repetition attempts.
- The maximum number of hours permitted for improvement for courses in which the student obtained points greater than or equal to 2.00 or higher is **9 hours**, whether for a single course or a number of courses.
- Their cumulative average is calculated on this basis, counting the course hours only once.
- All instances of repetition and the grade (or evaluation) obtained by the student in each repetition appear on the student's detailed transcript.

Article (18): Summer Training

- Summer training is organized for a period of **6 weeks**, with no less than 12 hours per week, provided that the student has successfully completed at least **70 credit hours** of the total credit hours required for graduation.
- A faculty member is assigned as responsible for the training, along with a number of assistant staff members, to follow up with the training participants annually and to assign evaluations for each of them according to the criteria determined by the College Council.

• **The summer training course is compulsory for all students and is not counted within the credit hours that enter into the calculation of the cumulative average. Rather, it is an acquired hours among the graduation requirements. Therefore, the student's grade in it is Pass/Fail, and it is mentioned in the student's detailed transcript.**

Article (19): Dismissal from the College

• A student who receives an academic probation in **4 consecutive regular semesters** or **6 separate regular semesters** (with the possibility that some may be consecutive) is dismissed.
• A student is dismissed from the college if they exceed the maximum duration of study at the college, after deducting the semesters of suspended enrollment.
• A student subject to dismissal for any of the aforementioned reasons may be given an additional and final opportunity to register for **two consecutive regular semesters** in addition to a summer semester to fulfill the graduation requirements, provided they have successfully completed no less than **105 hours** of the total hours required for graduation, and subject to the approval of the College Council and the University.

Article (20): Student Grievances

• Any student has the right to submit a request to appeal their exam result in one or more courses within a maximum period of one week from the date of the results announcement.
• The student is notified in writing of the result of the examination of their submitted grievance.

Article (21): Honors Rank and Student Ranking

• A student is granted honors rank if they successfully complete the courses they have studied at each academic level with a grade of no less than Very Good (i.e., a cumulative GPA CGPA of no less than **3.00**), provided that the period of study does not exceed **4 years** (eight regular semesters), and that the student has not failed or been denied entry in any course during their study at the college or the college from which they transferred (if any).
• Students are ranked based on the cumulative GPA (CGPA). In case of a tie, they are ranked according to the total sum of grades.

Article (22): University Regulations, Traditions, and Penalties for Violations

- Every violation of university laws, regulations, and traditions is considered a disciplinary offense, particularly:
    - Acts that disrupt the order of the college or university facilities.
    - Incitement to abstain from attending lessons, exams, or other university activities.
    - Any act contrary to proper conduct and behavior inside or outside the university.
    - Any violation of exam regulations, and any cheating in an exam or attempt thereof.
- Any student who commits cheating in an exam or attempts to do so and is caught in the act is removed by the Dean or the Head of the Examination Committee and is prohibited from taking the exam.
- Disciplinary penalties are:
    - Verbal or written warning.
    - Probation.
    - Deprivation of certain student services.
    - Deprivation from attending classes of one or more courses for a period not exceeding one month.
    - Dismissal from the college for a period not exceeding one month.
    - Deprivation from taking exams in one or more courses.
    - Dismissal from the college for a period not exceeding one semester.
    - Final dismissal from the university.
- The Dean of the college has the authority to impose the first six penalties after their approval by the University President.
- The Disciplinary Council has the authority to impose all penalties. The decision to refer a student to the Disciplinary Council is issued by the University President upon the request of the Dean.
- The Student Disciplinary Council is formed as follows:
    - The Dean of the College.
    - The Vice Dean of the College.
    - One of the faculty members of the college, chosen annually by the College Council.

Article (23): Departmental Coding System

The following table illustrates the coding system used for courses:

Code	Classification	#
	Basic Sciences العلوم الأساسية	1
BMT	▪ Mathimatics and Statistics (رياضيات و إحصاء)	
BPH	▪ Physics (فيزياء)	
	Humanities, Ethical and Social Sciences علوم أنسانية وأخلاقيه وإجتماعية	2
ENG	▪ لغة إنجليزية English	
PSC	▪ حقوق إنسان Human Rights	
PSY	▪ علم نفس Psychology	
SOC	▪ علم الإجتماع Sociology	
EPS	▪ ريادة الأعمال Entrepreneurship	
MGT	▪ إدارة Management	
CPS	▪ مهارات التواصل وكتابة التقارير Commincation Skills and Report Writing	
SCT	▪ التفكير العلمي Scientific Thinking	
CSE	▪ الأخلاق والآثار الاجتماعية للحوسبة Ethics and Social Impacts of Computing	
CS	**Computer Science Department Courses مقررات قسم علوم الحاسب**	3
CSC	▪ برنامج علوم الحاسب Computer Science Program	
CSA	▪ برنامج الذكاء الإصطناعي Artificial Intelligence Program	
CSS	▪ برنامج الأمن السيبراني Cybersecurity Program	
DM	**Digital Media Technology Department Courses مقررات قسم تكنولوجيا الوسائط الرقمية**	4
DMT	▪ برنامج تكنولوجيا الوسائط الرقمية Digital Media Technology Program	
IS	**Information Systems Department Courses مقررات قسم نظم المعلومات**	5
ISI	▪ برنامج نظم المعلومات Information Systems Program	
ISD	▪ برنامج علوم البيانات Data Science Program	
PR	**Graduation Project مشروع التخرج**	6
TR	**Summer Training التدريب الصيفي**	

الباب الثاني: هيكل البرامج والمقررات ومقترح الخطط الدراسية

برنامج علوم الحاسب – Computer Science Program

رسالة البرنامج

يلتزم برنامج علوم الحاسب كلية الحاسبات وتكنولوجيا المعلومات جامعة المستقبل بإعداد خريج متميز يواكب التطور التكنولوجي ومتطلبات سوق العمل قادر على المنافسة محليا وإقليميا والإبداع في تقديم حلول و أبحاث مبتكره لخدمة المجتمع والتنمية المستدامة وذلك في إطار القيم الجوهرية والأخلاقيات المهنية.

المهارات الخاصة بالبرنامج

بالإضافة الي المهارات العامة السالف ذكرها يكتسب خريج برنامج علوم الحاسب المهارات الاتيه:

- القدرة على تحليل وتصميم حلول للمشاكل باستخدام التقنيات القياسية للرياضيات والاحتمالات والإحصاء.
- اختيار المنهجيات والأساليب المناسبة لحل المشاكل المختلفة.
- ابتكار تصميم مناسب لحل المشاكل والذي يواكب التقدم في مجال علوم الحاسب و يناسب سوق العمل.
- القدرة على العمل بشكل مستقل وبالتعاون مع الآخرين.
- الإلمام بالقضايا المهنية والأخلاقية والقانونية والأخلاقية المتعلقة بعلوم الحاسب.
- ترسيخ الحاجة إلى التعلم مدى الحياة.

***يتم تحديث المهارات وفق مستجدات وإستقراء سوق العمل و NARS الصادر عن الهيئة القومية لضمان جودة التعليم والإعتماد.**

نسب توزيع مقررات البرنامج وتناسقها مع المعايير المرجعية القياسية لقطاع الحاسبات

	Subject Area	NARS %	CSC Program %	Number of Courses	Number of Credit Hours
A	Humanities, Ethical and Social Sciences (Univ. Req.)	8-10	9.09	5 (Compulsory) 1 (Elective)	10 (Compulsory) 2 (Elective)
B	Mathematics and Basic Sciences	16-18	18.18	8 (Compulsory)	24 (Compulsory)
C	Basic Computing Sciences (institution req.)	26-28	27.27	12 (Compulsory)	36 (Compulsory)
D	Applied Computing Sciences (specialization)	28-30	29.64	12 (Compulsory) 1 (Elective)	36 (Compulsory) 3 (Elective)
E	Training	3-5	2.27	1 (Compulsory)	3 (Compulsory)
F	Projects	3-5	4.55	2 (Compulsory)	6 (Compulsory)
	Subtotal	84-96	91	40 (Compulsory) 2 (Elective)	115 (Compulsory) 5 (Elective)
G	Optional (Institution character-identifying subjects)	16-4	9	4 (Elective)	12 (Elective)
	Total	100	100		132

A. Humanities, Ethical and Social Sciences (Univ. Req.) 12 C.H 9.09%

- أولًا: متطلبات إجبارية (10 ساعة معتمدة) **Compulsory (10 Credit Hours)**

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
ENG101	English - 1 لغة انجليزية -1	-	2	2	-	-
ENG102	English - 2 لغة انجليزية -2	ENG101 English -1	2	2	-	-
PSC110	Human Rights حقوق الانسان	-	2	2	-	-
CPS121	Communication Skills and Report Writing مهارات التواصل و كتابة التقارير	ENG101 English -1	2	-	4	-
CSE211	Ethics and Social Impacts of Computing الأخلاق والآثار الاجتماعية للحوسبة	-	2	2	-	-

- ثانياً: متطلبات إختيارية (2 ساعة معتمدة) **Electives (2 Credit Hours)**

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
PSY101	Psychology علم النفس	-	2	2	-	-
SOC101	Sociology علم الإجتماع	-	2	2	-	-
MGT103	Introduction to Management مبادئ إدارة	-	2	2	-	-
SCT101	Scientific Thinking التفكير العلمى	-	2	2	-	-
EPS101	Entrepreneurship ريادة الأعمال	-	2	2	-	-

B. Mathematics and Basic Sciences 24 C.H 18.18%

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
BMT111	Calculus تفاضل و تكامل	-	3	2	2	-
BMT121	Differential Equations معادلات تفاضليه	BMT111 Calculus	3	2	2	-
BMT212	Linear Algebra الجبر الخطي	BMT121 Differential Equations	3	2	2	-
BMT112	Discrete Mathematics تراكيب محددة	-	3	2	2	-
BPH111	Physics فيزياء	-	3	2	-	2
BPH211	Advanced Physics فيزياء متقدمه	BPH111 Physics	3	3	-	2
BMT122	Probability and Statistics إحصاء و إحتمالات	BMT111 Calculus	3	2	2	-
BMT211	Advanced Probability and Statistics إحصاء و احتمالات متقدمه	BMT122 Probability and Statistics	3	2	2	-

C. Basic Computing Sciences (institution req.) 36 C.H 27.27%

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
CSC111	Computing Fundamentals اساسيات الحاسب	-	3	2	1	2
CSC121	Structured Programming برمجه هيكلية	CSC111 Computing Fundamentals	3	2	1	2
CSC211	Object Oriented Programming برمجه شيئية	CSC121 Structured Programming	3	2	1	2
CSC221	Advanced Programming برمجة متقدمة	CSC211 Object Oriented Programming	3	2	1	2
CSC222	Data Structures هياكل البيانات	CSC211 Object Oriented Programming	3	2	1	2
CSC311	Operating Systems -1 نظم تشغيل -1	CSC211 Object Oriented Programming	3	2	-	2
CSC223	Software Engineering -1 هندسة البرمجيات -1	CSC211 Object Oriented Programming	3	2	-	2
CSC122	Logic Design التصميم المنطقي	BMT112 Discrete Mathematics	3	2	-	2
ISI222	Database Systems -1 نظم قواعد البيانات - 1	CSC121 Structured Programming	3	2	-	2
ISI311	Modeling and Simulation النمذجة والمحاكاة	BMT122 Probability and Statistics	3	2	-	2
CSS221	Computer Networks شبكات الحاسب	CSC111 Computing Fundamentals and BMT112 Discrete Mathematics	3	2	-	2
CSC312	Analysis and Design of Algorithms تحليل وتصميم الخوارزميات	CSC222 Data Structures	3	2	-	2

D. Applied Computing Sciences (specialisation) + Institution Character-Identifying subjects 51 C.H 38.64%

- أولًا: متطلبات إجبارية (36 ساعة معتمدة) **Compulsory (36 Credit Hours)**

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
CSC323	Computer Architecture عمارة الحاسب	CSC122 Logic Design	3	2	-	2
CSC313	Software Engineering - 2 هندسة البرمجيات - 2	CSC223 Software Engineering - 1	3	2	-	2
CSA311	Artificial Intelligence Fundamentals أساسيات الذكاء الإصطناعي	CSC211 Object Oriented Programming	3	2	-	2
CSC322	Operating Systems - 2 نظم التشغيل - 2	CSC311 Operating Systems - 1	3	2	-	2
CSC324	Theory of Computations نظرية الحسابات	BMT112 Discrete Mathematics	3	2	-	2
CSC325	Computer Organization and Assembly Language تنظيم الحاسب ولغة التجميع	CSC121 Structured Programming and CSC122 Logic Design	3	2	-	2
CSC412	Embedded Systems النظم المدمجة	CSC323 Computer Architecture	3	2	-	2
DMT323	Computer Graphics الرسم بالحاسب	BMT212 Linear Algebra	3	2	-	2
CSC422	Concepts of Programming Languages مفاهيم لغات البرمجة	CSC221 Advanced Programming	3	2	-	2
CSS312	Computers and Information Security أمن الحاسبات والمعلومات	CSS221 Computer Networks	3	2	-	2
CSC326	Web Applications Development تطوير تطبيقات الإنترنت (Lab Course مقررمعمل)	CSC211 Object Oriented Programming and ISI222 Database Systems -1	3	-	-	6

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
CSC411	Cloud Computing الحوسبة السحابية	CSC223 Software Engineering - 1 and CSS221 Computer Networks	3	2	-	2

- **ثانياً:** متطلبات إختيارية (15 ساعة معتمدة) **Electives (15 Credit Hours)**

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
CSC426	Mobile Applications Development تطوير تطبيقات المحمول (Lab Course مقررمعمل)	CSC211 Object Oriented Programming	3	-	-	6
CSA313	Image Processing معالجة الصور	BMT212 Linear Algebra and CSC121 Structured Programming	3	2	-	2
ISI324	Software Project Management إدارة مشاريع البرمجة	CSC223 Software Engineering -1	3	2	-	2
CSA324	Processing of Formal and Natural Language معالجة اللغات الرسمية و الطبيعية	CSA311 Artificial Intelligence Fundamentals	3	2	-	2
ISI427	Knowledge Based Systems نظم قواعد المعرفة	CSA311 Artificial Intelligence Fundamentals	3	2	-	2
CSC423	Distributed Processing المعالجة الموزعه	CSC323 Computer Architecture	3	2	-	2
ISI325	Operations Research بحوث عمليات	BMT212 Linear Algebra	3	2	-	2
CSC414	Selected Topics in Computer Science-1 موضوعات مختاره في علوم الحاسب -1	CSC211 Object Oriented Programming	3	2	-	2

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
CSC424	Selected Topics in Computer Science - 2 موضوعات مختاره في علوم الحاسب - 2	CSC211 Object Oriented Programming	3	2	-	2
CSA321	Machine Learning تعلم الآلة	BMT122 Probability and Statistics and CSA311 Artificial Intelligence Fundamentals	3	2	-	2
CSC415	Software Design & Architecture عمارة و تصميم البرمجيات	CSC223 Software Engineering - 1	3	2	-	2
CSC327	Software Testing إختبار البرمجيات	CSC223 Software Engineering - 1	3	2	-	2

E. Training 3 E.H 2.27%

Course Code كود المقرر	Course Name اسم المقرر	Earned Hours	Contact Hours ساعات الإتصال		
			Lecture	Tutorial	Practical
TR333	Summer Training التدريب الصيفي	3	لايقل عن 12 ساعة أسبوعيا لمدة ست أسابيع		

- يتم تسجيل الطالب في مقرر التدريب الصيفي بشرط إجتيازة 70 ساعة معتمدة على الأقل.

F. Projects 6 C.H 4.55%

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
PR498	Project -1 مشروع - 1	CSC223 Software Engineering - 1	3	1	-	4
PR499	Project - 2 مشروع - 2	PR498 Project -1	3	1	-	4

الفصل الدراس ي الأول

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
ENG101	English - 1 لغة انجليزية - 1	2	-
BMT111	Calculus تفاضل وتكامل	3	-
BMT112	Discrete Mathematics تراكيب محددة	3	-
BPH111	Physics الفيزياء	3	-
CSC111	Computing Fundamentals اساسيات الحوسبة	3	-
PSC110	Human Rights حقوق الإنسان	2	-
Total **المجمـوع**		**16**	

الفصل الدراس ي الثاني

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
ENG102	English - 2 لغة انجليزية - 2	2	ENG101 English -1
BMT121	Differential Equations معادلات تفاضليه	3	BMT111 Calculus
CSC121	Structured Programming برمجه هيكليه	3	CSC111 Computing Fundamentals
CPS121	Communication Skills and Report Writing مهارات التواصل وكتابة التقارير	2	ENG101 English -1
BMT122	Probability and Statistics إحصاء وإحتمالات	3	BMT111 Calculus
CSC122	Logic Design التصميم المنطقي	3	BMT112 Discrete Mathematics
Total **المجمـوع**		**16**	

الفصل الدراس ي الثالث

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
CPS211	Ethics and Social Impacts of Computing الأخلاق والآثار الاجتماعية للحوسبة	2	-
CSC211	Object Oriented Programming برمجه شيئية	3	CSC121 Structured Programming
BPH211	Advanced Physics الفيزياء المتقدمة	3	BPH111 Physics
BMT212	Linear Algebra الجبر الخطي	3	BMT121 Differential Equations
-	مقرر إختياري من (علوم أنسانية وأخلاقيه وإجتماعية)	2	-
BMT211	Advanced Probability and Statistics إحصاء وإحتمالات متقدمه	3	BMT122 Probability and Statistics
Total **المجمـوع**		**16**	

الفصل الدراس ي الرابع

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
CSC221	Advanced Programming برمجة متقدمة	3	CSC211 Object Oriented Programming
CSC222	Data Structures هياكل البيانات	3	CSC211 Object Oriented Programming
CSC223	Software Engineering -1 هندسة البرمجيات - 1	3	CSC211 Object Oriented Programming
CSS221	Computer Networks شبكات الحاسب	3	CSC111 Computing Fundamentals and BMT112 Discrete Mathematics
ISI222	Database Systems -1 نظم قواعد البيانات - 1	3	CSC121 Structured Programming
Total **المجمـوع**		**15**	

الفصل الدراس ي الخامس

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
ISI311	Modeling and Simulation النمذجة والمحاكاه	3	BMT122 Probability and Statistics
CSA311	Artificial Intelligence Fundamentals أساسيات الذكاء الإصطناعي	3	CSC211 Object Oriented Programming
CSS312	Computers and Information Security أمن الحاسبات والمعلومات	3	CSS221 Computer Networks
CSC311	Operating Systems - 1 نظم التشغيل - 1	3	Operating Systems -1 نظم تشغيل-1
CSC312	Analysis and Design of Algorithms تحليل وتصميم الخوارزميات	3	CSC222 Data Structures
CSC313	Software Engineering - 2 هندسة البرمجيات - 2	3	CSC223 Software Engineering - 1
Total **المجمـوع**		**18**	

الفصل الدراس ي السادس

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
CSC326	Web Applications Development تطوير تطبيقات الإنترنت	3	CSC211 Object Oriented Programming and ISI222 Database Systems -1
CSC322	Operating Systems - 2 نظم التشغيل - 2	3	CSC311 Operating Systems - 1
CSC323	Computer Architecture عمارة الحاسب	3	CSC122 Logic Design
CSC324	Theory of Computations نظرية الحسابات	3	BMT112 Discrete Mathematics
CSC325	Computer Organization and Assembly Language تنظيم الحاسب ولغة التجميع	3	CSC121 Structured Programming and CSC122 Logic Design
-	Computer Science Program Elective -1 إختياري برنامج علوم الحاسب - 1	3	*The prerequisite of the selected elective course
Total **المجمـوع**		**18**	

التدريب الصيفي

Course Code رقم المقرر	Course Name اسم المقرر	Earned Hours الساعات المكتسبة
TR333	Summer Training التدريب الصيفي	3

الفصل الدراس ي السابع

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
PR498	Project - 1 مشروع - 1	3	CSC223 Software Engineering - 1
CSC411	Cloud Computing الحوسبة السحابية	3	CSC223 Software Engineering - 1 and CSS221 Computer Networks
CSC412	Embedded Systems النظم المدمجة	3	CSC323 Computer Architecture
-	Computer Science Program Elective - 2 إختياري برنامج علوم الحاسب - 2	3	*The prerequisite of the selected elective course
-	Computer Science Program Elective - 3 إختياري برنامج علوم الحاسب - 3	3	*The prerequisite of the selected elective course
Total **المجمـوع**		**15**	

الفصل الدراس ي الثامن

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
PR499	Project - 2 مشروع - 2	3	PR498 Project -1
DMT323	Computer Graphics الرسم بالحاسب	3	BMT212 Linear Algebra
CSC422	Concepts of Programming Languages مفاهيم لغات البرمجة	3	CSC221 Advanced Programming
-	Computer Science Program Elective - 4 إختياري برنامج علوم الحاسب - 4	3	* The prerequisite of the selected elective course
-	Computer Science Program Elective - 5 إختياري برنامج علوم الحاسب - 5	3	*The prerequisite of the selected elective course
Total **المجمـوع**		**15**	

برنامج الذكاء الإصطناعي - Artificial Intelligence Program

رسالة البرنامج

يلتزم برنامج الذكاء الإصطناعي بكلية الحاسبات وتكنولوجيا المعلومات جامعة المستقبل بإعداد خريج متميز يواكب التطور التكنولوجي و قادر علي المنافسة في سوق العمل محليا وإقليميا ودوليا من خلال

تعليم يواكب المعايير الدولية، والمشاركة بفاعلية في مجال البحث العلمي لخدمة المجتمع وتحقيق اهداف التنمية المستدامة وذلك في إطار القيم الجوهرية والأخلاقيات المهنية.

المهارات الخاصة بالبرنامج

بالإضافة الي المهارات العامة السالف ذكرها يكتسب خريج برنامج الذكاء الإصطناعي المهارات الاتيه:

- القدرة على استخدام التقنيات القياسية للرياضيات ، والاحتمالات ، والإحصاء لمعالجة المشاكل النموذجية لمهنة في مجال الذكاء الاصطناعي.
- استخدام حزم البرامج التي تدعم تطبيقات تعلم الآلة على أساس البيانات ذات الجودة التي تُعطى للآلة القدرة علي حل المشاكل في مختلف طبقات الذكاء الاصطناعي.
- تعرف على الموارد القياسية التي يمكن استخدامها للبقاء على اطلاع دائم بقضايا الذكاء الاصطناعي والممارسات والتطورات المهنية الحالية.
- تلخيص وتوصيل مشاكل الذكاء الاصطناعي شفهياً وخطياً إلى الجماهير المتخصصة وغير المتخصصة. - القدرة على العمل بشكل مستقل وبالتعاون مع الآخرين.
- تطبيق معايير عالية للسلوك الأخلاقي والقدرة على التعبير عن أفضل ممارسات العمل الأخلاقي في الممارسة المهنية.
- ترسيخ الحاجة إلى التعلم مدى الحياة.

***يتم تحديث المهارات وفق مستجدات وإستقراء سوق العمل و NARS الصادر عن الهيئة القومية لضمان جودة التعليم والإعتماد.**

نسب توزيع مقررات البرنامج وتناسقها مع المعايير المرجعية القياسية لقطاع الحاسبات

	Subject Area	NARS %	CSA Program %	Number of Courses	Number of Credit Hours
A	Humanities, Ethical and Social Sciences (Univ. Req.)	8-10	9.09	5 (Compulsory) 1 (Elective)	10 (Compulsory) 2 (Elective)
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E	Training	3-5	2.27	1 (Compulsory)	3 (Compulsory)
F	Projects	3-5	4.55	2 (Compulsory)	6 (Compulsory)
	Subtotal	84-96	91	40 (Compulsory) 2 (Elective)	115 (Compulsory) 5 (Elective)
G	Optional (Institution character-identifying subjects)	16-4	9	4 (Elective)	12 (Elective)
	Total	100	100		132

A. Humanities, Ethical and Social Sciences (Univ. Req.) 12 C.H 9.09%

• أولًا: متطلبات إجبارية (10 ساعة معتمدة) **Compulsory (10 Credit Hours)**

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
ENG101	English -1 لغة انجليزية - 1	-	2	2	-	-
ENG102	English - 2 لغة انجليزية - 2	ENG101 English - 1	2	2	-	-
PSC110	Human Rights حقوق الانسان	-	2	2	-	-
CPS121	Communication Skills and Report Writing مهارات التواصل و كتابة التقارير	ENG101 English - 1	2	-	4	-
CSE211	Ethics and Social Impacts of Computing الأخلاق والآثار الاجتماعية للحوسبة	-	2	2	-	-

• ثانياً: متطلبات إختيارية (2 ساعة معتمدة) **Electives (2 Credit Hours)**

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
PSY101	Psychology علم النفس	-	2	2	-	-
SOC101	Sociology علم الإجتماع	-	2	2	-	-
MGT103	Introduction to Management مبادئ إدارة	-	2	2	-	-
SCT101	Scientific Thinking التفكير العلمى	-	2	2	-	-
EPS101	Entrepreneurship ريادة الأعمال	-	2	2	-	-

B. Mathematics and Basic Sciences 24 C.H 18.18%

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
BMT111	Calculus تفاضل و تكامل	-	3	2	2	-
BMT121	Differential Equations معادلات تفاضليه	BMT111 Calculus	3	2	2	-
BMT212	Linear Algebra الجبر الخطي	BMT121 Differential Equations	3	2	2	-
BMT112	Discrete Mathematics تراكيب محددة	-	3	2	2	-
BPH111	Physics فيزياء	-	3	2	-	2
BPH211	Advanced Physics فيزياء متقدمه	BPH111 Physics	3	3	-	2
BMT122	Probability and Statistics إحصاء و إحتمالات	BMT111 Calculus	3	2	2	-
BMT211	Advanced Probability and Statistics إحصاء و احتمالات متقدمه	BMT122 Probability and Statistics	3	2	2	-

C. Basic Computing Sciences (institution req.) 36 C.H 27.27%

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
CSC111	Computing Fundamentals اساسيات الحاسب	-	3	2	1	2
CSC121	Structured Programming برمجه هيكلية	CSC111 Computing Fundamentals	3	2	1	2
CSC211	Object Oriented Programming برمجه شيئية	CSC121 Structured Programming	3	2	1	2
CSC221	Advanced Programming برمجة متقدمة	CSC211 Object Oriented Programming	3	2	1	2
CSC222	Data Structures هياكل البيانات	CSC211 Object Oriented Programming	3	2	1	2
CSC311	Operating Systems - 1 نظم تشغيل - 1	CSC211 Object Oriented Programming	3	2	-	2
CSC223	Software Engineering - 1 هندسة البرمجيات - 1	CSC211 Object Oriented Programming	3	2	-	2
CSC122	Logic Design التصميم المنطقي	BMT112 Discrete Mathematics	3	2	-	2
ISI222	Database Systems - 1 نظم قواعد البيانات - 1	CSC121 Structured Programming	3	2	-	2
ISI311	Modeling and Simulation النمذجة والمحاكاة	BMT122 Probability and Statistics	3	2	-	2
CSS221	Computer Networks شبكات الحاسب	CSC111 Computing Fundamentals and BMT112 Discrete Mathematics	3	2	-	2
CSC312	Analysis and Design of Algorithms تحليل وتصميم الخوارزميات	CSC222 Data Structures	3	2	-	2

D. Applied Computing Sciences (specialisation) + Institution Character-Identifying subjects 51 C.H 38.64%

Compulsory (39 Credit Hours) أولًا: متطلبات إجبارية (39 ساعة معتمدة)

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
CSC323	Computer Architecture عمارة الحاسب	CSC122 Logic Design	3	2	-	2
CSA322	Reasoning and Agents استنتاج وتمثيل المعرفة	CSA311 Artificial Intelligence Fundamentals	3	2	-	2
CSA311	Artificial Intelligence Fundamentals أساسيات الذكاء الإصطناعي	CSC211 Object Oriented Programming	3	2	-	2
CSA411	Neural Networks and Deep Learning الشبكات العصبية والتعلم العميق	BMT121 Differential Equations and CSA311 Artificial Intelligence Fundamentals	3	2	-	2
CSA313	Image Processing معالجة الصور	BMT212 Linear Algebra and CSC121 Structured Programming	3	2	-	2
CSA323	Artificial Intelligence Applications تطبيقات الذكاء الإصطناعي	CSA311 Artificial Intelligence Fundamentals	3	2	-	2
CSA324	Processing of Formal and Natural Language معالجة اللغات الرسمية و الطبيعية	CSA311 Artificial Intelligence Fundamentals	3	2	-	2
CSC421	Embedded Systems النظم المدمجة	CSC323 Computer Architecture	3	2	-	2
CSA321	Machine learning تعلم الآلة	BMT122 Probability and Statistics and CSA311	3	2	-	2

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
		Artificial Intelligence Fundamentals				
CSA422	Intelligent Autonomous Robotics الإنسان الآلي ذاتي التحكم الذكي	CSC421 Embedded Systems	3	2	-	2
CSS312	Computers and Information Security أمن الحاسبات والمعلومات	CSS221 Computer Networks	3	2	-	2
CSA423	Computer Vision الرؤية بالحاسب	CSA313 Image Processing and CSA311 Artificial Intelligence Fundamentals	3	2	-	2
CSA424	Reinforcement Learning التعلم التعزيزي	CSA321 Machine Learning	3	2	-	2

ثانياً: متطلبات إختيارية (12 ساعة معتمدة) **Electives (12 Credit Hours)**

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
CSA417	Bioinformatics Systems نظم المعلومات الحيوية	BMT122 Probability and Statistics	3	2	-	2
CSC313	Software Engineering - 2 هندسة البرمجيات - 2	CSC223 Software Engineering - 1	3	2	-	2
CSC411	Cloud Computing الحوسبة السحابية	CSC223 Software Engineering - 1 and CSS221 Computer Networks	3	2	-	2
CSA412	Artificial Intelligence for Cyber Security الذكاء الإصطناعى في الأمن السيبراني	CSA311 Artificial Intelligence Fundamentals	3	2	-	2

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
CSA413	Pattern Recognition التعرف على الأنماط	CSA313 Image Processing and CSA321 Machine Learning	3	2	-	2
CSC426	Mobile Applications Development تطوير تطبيقات المحمول (Lab Course مقررمعمل)	CSC211 Object Oriented Programming	3	-	-	6
CSA426	Data Analytics تحليل البيانات	BMT122 Probability and Statistics	3	2	-	2
CSA416	Soft Computing الحوسبة المرنة	CSA311 Artificial Intelligence Fundamentals	3	2	-	2
CSA425	Computing Cognitive Sciences العلوم الادراكية الحسابية	CSA322 Reasoning and Agents	3	2	-	2
CSA415	Selected Topics in Artificial Intelligence - 1 موضوعات مختاره في الذكاء الإصطناعى - 1	CSA311 Artificial Intelligence Fundamentals	3	2	-	2
CSA428	Selected Topics in Artificial Intelligence - 2 موضوعات مختاره في الذكاء الإصطناعى - 2	CSA311 Artificial Intelligence Fundamentals	3	2	-	2
CSC326	Web Applications Development تطوير تطبيقات الإنترنت (Lab Course مقررمعمل)	CSC211 Object Oriented Programming and ISI222 Database Systems -1	3	-	-	6

E. Training 4 E.H 2.27%

Course Code كود المقرر	Course Name اسم المقرر	Earned Hours	Contact Hours ساعات الإتصال		
			Lecture	Tutorial	Practical
TR333	Summer Training التدريب الصيفي	3	لايقل عن 12 ساعة أسبوعيا لمدة ست أسابيع		

- يتم تسجيل الطالب في مقرر التدريب الصيفي بشرط إجتيازة 70 ساعة معتمدة على الأقل.

F. Projects 6 C.H 4.55%

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
PR498	Project - 1 مشروع - 1	CSC223 Software Engineering -1	3	1	-	4
PR499	Project - 2 مشروع - 2	PR498 Project -1	3	1	-	4

الفصل الدراس ي الأول

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
ENG101	English - 1 لغة انجليزية - 1	2	-
BMT111	Calculus تفاضل وتكامل	3	-
BMT112	Discrete Mathematics تراكيب محددة	3	-
BPH111	Physics الفيزياء	3	-
CSC111	Computing Fundamentals اساسيات الحوسبة	3	-
PSC110	Human Rights حقوق الإنسان	2	-
Total **المجمـوع**		**16**	

الفصل الدراس ي الثاني

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
ENG102	English - 2 لغة انجليزية - 2	2	ENG101 English -1
BMT121	Differential Equations معادلات تفاضليه	3	BMT111 Calculus
CSC121	Structured Programming برمجه هيكليه	3	CSC111 Computing Fundamentals
CPS121	Communication Skills and Report Writing مهارات التواصل وكتابة التقارير	2	ENG101 English -1
BMT122	Probability and Statistics إحصاء وإحتمالات	3	BMT111 Calculus
CSC122	Logic Design التصميم المنطقي	3	BMT112 Discrete Mathematics
Total **المجمـوع**		**16**	

الفصل الدراس ي الثالث

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
CPS211	Ethics and Social Impacts of Computing الأخلاق والآثار الاجتماعية للحوسبة	2	-
CSC211	Object Oriented Programming برمجه شيئية	3	CSC121 Structured Programming
BPH211	Advanced Physics الفيزياء المتقدمة	3	BPH111 Physics
BMT212	Linear Algebra الجبر الخطي	3	BMT121 Differential Equations
-	مقرر إختياري من (علوم أنسانية وأخلاقيه وإجتماعية)	2	-
BMT211	Advanced Probability and Statistics إحصاء وإحتمالات متقدمه	3	BMT122 Probability and Statistics
Total **المجمـوع**		**16**	

الفصل الدراس ي الرابع

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
CSC221	Advanced Programming برمجة متقدمة	3	CSC211 Object Oriented Programming
CSC222	Data Structures هياكل البيانات	3	CSC211 Object Oriented Programming
CSC223	Software Engineering -1 هندسة البرمجيات - 1	3	CSC211 Object Oriented Programming
CSS221	Computer Networks شبكات الحاسب	3	CSC111 Computing Fundamentals and BMT112 Discrete Mathematics
ISI222	Database Systems -1 نظم قواعد البيانات - 1	3	CSC121 Structured Programming
Total **المجمـوع**		**15**	

الفصل الدراس ي الخامس

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
ISI311	Modeling and Simulation النمذجة والمحاكاه	3	BMT122 Probability and Statistics
CSA311	Artificial Intelligence Fundamentals أساسيات الذكاء الإصطناعي	3	CSC211 Object Oriented Programming
CSS312	Computers and Information Security أمن الحاسبات والمعلومات	3	CSS221 Computer Networks
CSC311	Operating Systems - 1 نظم التشغيل - 1	3	CSC211 Object Oriented Programming
CSC312	Analysis and Design of Algorithms تحليل وتصميم الخوارزميات	3	CSC222 Data Structures
CSA313	Image Processing معالجة الصور	3	BMT212 Linear Algebra and CSC121 Structured Programming
Total المجمـوع		**18**	

الفصل الدراس ي السادس

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
CSC323	Computer Architecture عمارة الحاسب	3	CSC122 Logic Design
CSA321	Machine learning تعلم الآلة	3	BMT122 Probability and Statistics and CSA311 Artificial Intelligence Fundamentals
CSA322	Reasoning and Agents استنتاج وتمثيل المعرفة	3	CSA311 Artificial Intelligence Fundamentals
CSA423	Computer Vision الرؤية بالحاسب	3	CSA313 Image Processing And CSA311 Artificial Intelligence Fundamentals
CSA323	Artificial Intelligence Applications تطبيقات الذكاء الإصطناعي	3	CSA311 Artificial Intelligence Fundamentals
CSA324	Processing of Formal and Natural Language معالجة اللغات الرسمية و الطبيعية	3	CSA311 Artificial Intelligence Fundamentals
Total - المجموع		**18**	

التدريب الصيفي

Course Code رقم المقرر	Course Name اسم المقرر	Earned Hours الساعات المكتسبة
TR333	Summer Training التدريب الصيفي	3

الفصل الدراس ي السابع

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
PR498	Project - 1 مشروع - 1	3	CSC223 Software Engineering -1
CSA411	Neural Networks and Deep Learning الشبكات العصبية والتعلم العميق	3	BMT121 Differential Equations and CSA311 Artificial Intelligence Fundamentals
CSC412	Embedded Systems النظم المدمجة	3	CSC323 Computer Architecture
-	AI Program Elective - 1 إختياري برنامج الذكاء الإصطناعي - 1	3	*The prerequisite of the selected elective course
-	AI Program Elective - 2 إختياري برنامج الذكاء الإصطناعي - 2	3	*The prerequisite of the selected elective course
Total **المجمـوع**		**15**	

الفصل الدراس ي الثامن

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
PR499	Project - 2 مشروع - 2	3	Project -1 PR498
CSA424	Reinforcement Learning التعلم التعزيزي	3	CSA321 Machine Learning
CSA422	Intelligent Autonomous Robotics الانسان الالي ذاتي التحكم الذكي	3	CSC421 Embedded Systems
-	AI Program Elective - 3 إختياري برنامج الذكاء الإصطناعي - 3	3	*The prerequisite of the selected elective course
-	AI Program Elective - 4 إختياري برنامج الذكاء الإصطناعي - 4	3	*The prerequisite of the selected elective course
Total - المجموع		**15**	

برنامج الأمن السيبراني - Cybersecurity Program

رسالة البرنامج

يلتزم برنامج الامن السيبراني كلية الحاسبات وتكنولوجيا المعلومات جامعة المستقبل بإعداد خريجين على قدر كافٍ من التأهيل العلمي والتطبيقي مزودين بالمعرفة والمهارات والقيم ولهم مكانة متميزة في مجال الأمن السيبراني على المستوي المحلي والاقليمي والدولي والمساهمة في انتاج الابحاث العلمية المتميزة التي تخدم المجتمع واهداف التنمية المستدامة.

المهارات الخاصة بالبرنامج

بالإضافة الي المهارات العامة السالف ذكرها يكتسب خريج برنامج الأمن السيبراني المهارات الاتيه:

- القدرة على تحليل مشكلة أمنية محددة على نطاق واسع وتطبيق مبادئ الأمن السيبراني لتصميم وتنفيذ الحلول.
- الالتزام بالمبادئ والممارسات الأمنية للحفاظ على العمليات في ظل وجود مخاطر وتهديدات.
- التعرف على المسؤوليات المهنية واتخاذ أحكام مستنيرة في ممارسة الأمن السيبراني بناءً على المبادئ القانونية والأخلاقية.
- العمل بفعالية كعضو في فريق يشارك في أنشطة الأمن السيبراني.
- القدرة على تطبيق مبادئ وممارسات الأمان على المكونات البيئية والأجهزة والبرمجيات والشبكات والموارد البشرية داخل المنظومة التي يعمل بها.

***يتم تحديث المهارات وفق مستجدات وإستقراء سوق العمل و NARS الصادر عن الهيئة القومية لضمان جودة التعليم والإعتماد.**

نسب توزيع مقررات البرنامج وتناسقها مع المعايير المرجعية القياسية لقطاع الحاسبات

	Subject Area	NARS %	CSS Program %	Number of Courses	Number of Credit Hours
A	Humanities, Ethical and Social Sciences (Univ. Req.)	8-10	9.09	5 (Compulsory) 1 (Elective)	10 (Compulsory) 2 (Elective)
B	Mathematics and Basic Sciences	16-18	18.18	8 (Compulsory)	24 (Compulsory)
C	Basic Computing Sciences (institution req.)	26-28	27.27	12 (Compulsory)	36 (Compulsory)
D	Applied Computing Sciences (specialization)	28-30	29.64	12 (Compulsory) 1 (Elective)	36 (Compulsory) 3 (Elective)
E	Training	3-5	2.27	1 (Compulsory)	3 (Compulsory)
F	Projects	3-5	4.55	2 (Compulsory)	6 (Compulsory)
	Subtotal	84-96	91	40 (Compulsory) 2 (Elective)	115 (Compulsory) 5 (Elective)
G	Optional (Institution character-identifying subjects)	16-4	9	4 (Elective)	12 (Elective)
	Total	100	100		132

A. Humanities, Ethical and Social Sciences (Univ. Req.) 12 C.H 9.09%

- أولًا: متطلبات إجبارية (10 ساعة معتمدة) **Compulsory (10 Credit Hours)**

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
ENG101	English - 1 لغة انجليزية - 1	-	2	2	-	-
ENG102	English - 2 لغة انجليزية - 2	ENG101 English - 1	2	2	-	-
PSC110	Human Rights حقوق الانسان	-	2	2	-	-
CPS121	Communication Skills and Report Writing مهارات التواصل و كتابة التقارير	ENG101 English - 1	2	-	4	-
CSE211	Ethics and Social Impacts of Computing الأخلاق والآثار الاجتماعية للحوسبة	-	2	2	-	-

- ثانياً: متطلبات إختيارية (2 ساعة معتمدة) **Electives (2 Credit Hours)**

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
PSY101	Psychology علم النفس	-	2	2	-	-
SOC101	Sociology علم الإجتماع	-	2	2	-	-
MGT103	Introduction to Management مبادئ إدارة	-	2	2	-	-
SCT101	Scientific Thinking التفكير العلمى	-	2	2	-	-
EPS101	Entrepreneurship ريادة الأعمال	-	2	2	-	-

B. Mathematics and Basic Sciences 24 C.H 18.18%

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
BMT111	Calculus تفاضل و تكامل	-	3	2	2	-
BMT121	Differential Equations معادلات تفاضليه	BMT111 Calculus	3	2	2	-
BMT212	Linear Algebra الجبر الخطي	BMT121 Differential Equations	3	2	2	-
BMT112	Discrete Mathematics تراكيب محددة	-	3	2	2	-
BPH111	Physics فيزياء	-	3	2	-	2
BMT122	Probability and Statistics إحصاء و إحتمالات	BMT111 Calculus	3	2	2	-
BMT211	Advanced Probability and Statistics إحصاء و احتمالات متقدمه	BMT122 Probability and Statistics	3	2	2	-
BMT213	Number Theory نظرية الأعداد	BMT112 Discrete Mathematics	3	2	2	-

C. Basic Computing Sciences (institution req.) 36 C.H 27.27%

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
CSC111	Computing Fundamentals اساسيات الحاسب	-	3	2	1	2
CSC121	Structured Programming برمجه هيكلية	CSC111 Computing Fundamentals	3	2	1	2
CSC211	Object Oriented Programming برمجه شيئية	CSC121 Structured Programming	3	2	1	2

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
CSC221	Advanced Programming برمجة متقدمة	CSC211 Object Oriented Programming	3	2	1	2
CSC222	Data Structures هياكل البيانات	CSC211 Object Oriented Programming	3	2	1	2
CSC311	Operating Systems - 1 نظم تشغيل - 1	CSC211 Object Oriented Programming	3	2	-	2
CSC223	Software Engineering - 1 هندسة البرمجيات - 1	CSC211 Object Oriented Programming	3	2	-	2
CSC122	Logic Design التصميم المنطقي	BMT112 Discrete Mathematics	3	2	-	2
ISI222	Database Systems -1 نظم قواعد البيانات - 1	CSC121 Structured Programming	3	2	-	2
CSS221	Computer Networks شبكات الحاسب	CSC111 Computing Fundamentals and BMT112 Discrete Mathematics	3	2	-	2
CSS311	Cyber Security Fundamentals أساسيات الأمن السيبراني	CSS221 Computer Networks	3	2	-	2
CSC312	Analysis and Design of Algorithms تحليل وتصميم الخوارزميات	CSC222 Data Structures	3	2	-	2

D. Applied Computing Sciences (specialisation) + Institution Character-Identifying subjects 51 C.H 38.64%

Compulsory (36 Credit Hours) أولًا: متطلبات إجبارية (36 ساعة معتمدة)

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
CSS312	Computers and Information Security أمن الحاسبات والمعلومات	BMT112 Discrete Mathematics and CSS221 Computer Networks	3	2	-	2
CSS313	Advanced Computer Networks شبكات الحاسب المتقدمة	CSS221 Computer Networks	3	2	-	2
CSS321	Cryptography and Network Security التشفير و أمن الشبكات	BMT213 Number Theory and CSS312 Advanced Computer Networks	3	2	-	2
CSS322	Software Security أمن البرمجيات	CSC221 Advanced Programming and CSS312 Computers and Information Security	3	2	-	2
CSS323	Security Threats and Risk Analysis التهديدات الأمنية وتحليل المخاطر	CSS311 Cyber Security Fundamentals	3	2	-	2
CSS324	Ethical Hacking and Penetration Testing القرصنة الأخلاقية واختبار الإختراق	CSS313 Advanced Computer Networks	3	2	-	2
CSS411	Digital Forensics التحاليل الجنائية الرقمية	CSS322 Software Security	3	2	-	2

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
CSC411	Cloud Computing الحوسبة السحابية	CSC223 Software Engineering - 1 and CSS221 Computer Networks	3	2	-	2
CSS421	Mobile and Wireless Security أمن المحمول واللاسلكي	CSS321 Cryptography and Network Security	3	2	-	2
CSS422	Cyber Security Incident Detection and Response كشف حوادث الأمن السيبراني والاستجابة لها	CSS323 Security Threats and Risk Analysis	3	2	-	2
CSA311	Artificial Intelligence Fundamentals أساسيات الذكاء الإصطناعي	CSC211 Object Oriented Programming	3	2	-	2
CSC326	Web Applications Development تطوير تطبيقات الإنترنت (مقرر معمل Course	CSC211 Object Oriented Programming and ISI222 Database Systems -1	3	-	-	6

ثانياً: متطلبات إختيارية (15 ساعة معتمدة) Electives (15 Credit Hours)

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
CSC323	Computer Architecture عمارة الحاسب	CSC122 Logic Design	3	2	-	2
CSS325	Internet of Things (IoT) إنترنت الأشياء	CSS313 Advanced Computer Networks	3	2	-	2
CSS326	Cybersecurity Law and Policy قانون وسياسة الأمن السيبراني	CSS311 Cyber Security Fundamentals	3	2	-	2
CSS327	Cyber-Physical Security الأمن السيبراني المادي	CSS311 Cyber Security Fundamentals	3	2	-	2
CSS423	Cloud Security أمن الحوسبة السحابية	CSC411 Cloud Computing	3	2	-	2
CSS412	Web Application Security and Testing أمن تطبيقات الويب واختباره	CSC326 Web Applications Development	3	2	-	2
CSS413	Advanced Penetration Testing اختبارات الاختراق المتقدمة	CSS324 Ethical Hacking and Penetration Testing	3	2	-	2
CSA412	Artificial Intelligence for Cyber Security الذكاء الإصطناعى في الأمن السيبراني	CSA311 Artificial Intelligence Fundamentals	3	2	-	2
CSS425	Network Forensics التحليل الجنائي للشبكات	CSS321 Cryptography and Network Security	3	2	-	2
CSS426	Cyber Defense الدفاع السيبراني	CSS321 Cryptography and Network Security and CSS322 Software Security	3	2	-	2
CSS427	Malicious Software البرمجيات الخبيثة	CSS322 Software Security	3	2	-	2

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
ISD415	Introduction to Block Chain & Distributed Ledgers مقدمة في تقنية سلسلة الكتل و الحسابات الموزّعة	CSS312 Computers and Information Security	3	2	-	2
CSS428	Selected Topics in Cyber Security - 1 موضوعات مختاره في الأمن السيبراني - 1	CSS311 Cyber Security Fundamentals	3	2	-	2
CSS429	Selected Topics in Cyber Security - 2 موضوعات مختاره في الأمن السيبراني - 2	CSS311 Cyber Security Fundamentals	3	2	-	2

E. Training 3 E.H 2.27%

Course Code كود المقرر	Course Name اسم المقرر	Earned Hours	Contact Hours ساعات الإتصال		
			Lecture	Tutorial	Practical
TR333	Summer Training التدريب الصيفي	3	لايقل عن 12 ساعة أسبوعيا لمدة ست أسابيع		

- يتم تسجيل الطالب في مقرر التدريب الصيفي بشرط إجتيازة 70 ساعة معتمدة على الأقل.

F. Projects 6 C.H 5%

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
PR498	Project - 1 مشروع - 1	CSC223 Software Engineering -1	3	1	-	4
PR499	Project - 2 مشروع - 2	PR498 Project -1	3	1	-	4

الفصل الدراس ي الأول

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
ENG101	English - 1 لغة انجليزية - 1	2	-
BMT111	Calculus تفاضل وتكامل	3	-
BMT112	Discrete Mathematics تراكيب محددة	3	-
BPH111	Physics الفيزياء	3	-
CSC111	Computing Fundamentals اساسيات الحوسبة	3	-
PSC110	Human Rights حقوق الإنسان	2	-
Total **المجمـوع**		**16**	

الفصل الدراس ي الثاني

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
ENG102	English - 2 لغة انجليزية - 2	2	ENG101 English -1
BMT121	Differential Equations معادلات تفاضليه	3	BMT111 Calculus
CSC121	Structured Programming برمجه هيكليه	3	CSC111 Computing Fundamentals
CPS121	Communication Skills and Report Writing مهارات التواصل وكتابة التقارير	2	ENG101 English -1
BMT122	Probability and Statistics إحصاء وإحتمالات	3	BMT111 Calculus
CSC122	Logic Design التصميم المنطقي	3	BMT112 Discrete Mathematics
Total **المجمـوع**		**16**	

الفصل الدراس ي الثالث

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
CPS211	Ethics and Social Impacts of Computing الأخلاق والآثار الاجتماعية للحوسبة	2	-
CSC211	Object Oriented Programming برمجه شيئية	3	CSC121 Structured Programming
BMT213	Number theory نظرية الأعداد	3	BMT112 Discrete Mathematics
BMT212	Linear Algebra الجبر الخطي	3	BMT121 Differential Equations
-	مقرر إختياري من (علوم أنسانية وأخلاقيه وإجتماعية)	2	-
BMT211	Advanced Probability and Statistics إحصاء وإحتمالات متقدمه	3	BMT122 Probability and Statistics
Total **المجمـوع**		**16**	

الفصل الدراس ي الرابع

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
CSC221	Advanced Programming برمجة متقدمة	3	CSC211 Object Oriented Programming
CSC222	Data Structures هياكل البيانات	3	CSC211 Object Oriented Programming
CSC223	Software Engineering -1 هندسة البرمجيات - 1	3	CSC211 Object Oriented Programming
CSS221	Computer Networks شبكات الحاسب	3	CSC111 Computing Fundamentals and BMT112 Discrete Mathematics
ISI222	Database Systems -1 نظم قواعد البيانات - 1	3	CSC121 Structured Programming
Total **المجمـوع**		**15**	

الفصل الدراس ي الخامس

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
CSS311	Cyber Security Fundamentals أساسيات الأمن السيبراني	3	CSS221 Computer Networks
CSA311	Artificial Intelligence Fundamentals أساسيات الذكاء الإصطناعي	3	CSC211 Object Oriented Programming
CSS312	Computers and Information Security أمن الحاسبات والمعلومات	3	CSS221 Computer Networks
CSC311	Operating Systems - 1 نظم التشغيل - 1	3	CSC211 Object Oriented Programming
CSS313	Advanced Computer Networks شبكات الحاسب المتقدمة	3	CSS221 Computer Networks
CSC312	Analysis and Design of Algorithms تحليل وتصميم الخوارزميات	3	CSC222 Data Structures
Total **المجمـوع**		**18**	

الفصل الدراس ي السادس

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
CSS321	Cryptography and Network Security التشفير و أمن الشبكات	3	BMT213 Number Theory And CSS312 Advanced Computer Networks
CSC326	Web Applications Development تطوير تطبيقات الإنترنت	3	CSC211 Object Oriented Programming And ISI222 Database Systems -1
CSS322	Software Security أمن البرمجيات	3	CSC221 Advanced Programming and CSS312 Computers and Information Security
CSS323	Security Threats and Risk Analysis التهديدات الأمنية وتحليل المخاطر	3	CSS311 Cyber Security Fundamentals
CSS324	Ethical Hacking and Penetration Testing القرصنة الأخلاقية واختبار الإختراق	3	CSS313 Advanced Computer Networks
-	Cyber Security Program Elective - 1 اختياري برنامج الأمن السيبراني - 1	3	*The prerequisite of the selected elective course
Total **المجمـوع**		**18**	

التدريب الصيفي

Course Code رقم المقرر	Course Name اسم المقرر	Earned Hours الساعات المكتسبة
TR333	Summer Training التدريب الصيفي	3

الفصل الدراس ي السابع

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
PR498	Project - 1 مشروع - 1	3	CSC223 Software Engineering -1
CSC411	Cloud Computing الحوسبة السحابية	3	CSC223 Software Engineering - 1 And CSS221 Computer Networks
CSS411	Digital Forensics التحاليل الجنائية الرقمية	3	CSS322 Software Security
-	Cyber Security Program Elective - 2 إختياري برنامج الأمن السيبراني - 2	3	*The prerequisite of the selected elective course
-	Cyber Security Program Elective - 3 إختياري برنامج الأمن السيبراني - 3	3	*The prerequisite of the selected elective course
Total **المجموع**		**15**	

الفصل الدراس ي الثامن

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
PR499	Project - 2 مشروع - 2	3	PR498 Project -1
CSS421	Mobile and Wireless Security أمن المحمول واللاسلكي	3	CSS321 Cryptography and Network Security
CSS422	Cyber Security Incident Detection and Response كشف حوادث الأمن السيبراني والاستجابة لها	3	CSS323 Security Threats and Risk Analysis
-	Cyber Security Program Elective - 4 إختياري برنامج الأمن السيبراني - 4	3	*The prerequisite of the selected elective course
-	Cyber Security Program Elective - 5 إختياري برنامج الأمن السيبراني - 5	3	*The prerequisite of the selected elective course
Total - المجموع		**15**	

Information Systems Program - برنامج نظم المعلومات

رسالة البرنامج

يلتزم برنامج نظم المعلومات كلية الحاسبات وتكنولوجيا المعلومات جامعة المستقبل بتوفير بيئة تعليمية مناسبة تشجع على التعلم الذاتي والجماعي والمساهمة في إعداد خريج قادر على المنافسة محليا وإقليميا ودوليا وملتزما باخلاقيات المهنة والابتكار في البحث العلمي بما يساهم في إثراء المعرفة في مجال نظم المعلومات والتنمية المستدامة للمجتمع.

المهارات الخاصة بالبرنامج

بالإضافة الي المهارات العامة السالف ذكرها يكتسب خريج برنامج نظم المعلومات المهارات الاتيه:

- التعرف على الأساليب النوعية والكمية المختلفة لتحليل البيانات. - شرح أساسيات تصاميم تخطيط نظم المعلومات المختلفة.
- تصميم وتنفيذ قواعد البيانات.
- وصف وتطبيق تقنيات دعم القرار. - وصف أنظمة التجارة الإلكترونية. - شرح تقنيات التنقيب عن البيانات.
- وصف أنظمة إدارة قواعد البيانات (DBMS).
- تحديد الأفكار الرئيسية والأنماط والمكونات والسمات واكتشاف العلاقات بين هذه المكونات في تحليل البرامج مع تصميمات مختلفة.
- تحليل مشاكل نظم المعلومات المختلفة وتحديد الأهداف والمتطلبات.
- المقارنة والتمييز بين طرق التحليل والتقنيات المستخدمة في حلول مشاكل نظم المعلومات. - اختيار المنهجيات والتقنيات المناسبة لحل مشكلة معينة وتحديد حدودها وأخطاءها.
- تحليل المعلومات المتعلقة بحل المشكلة. - تحليل وضبط أنظمة قواعد البيانات.
- تصميم التفاعلات بين الإنسان والحاسوب.
- تصميم خطة فعالة لإدارة المخاطر واكتشاف جوانب السلامة. - تنفيذ تقنيات الإدارة المختلفة.
- تطبيق منهجيات نظم المعلومات المختلفة.

***يتم تحديث المهارات وفق مستجدات وإستقراء سوق العمل و NARS الصادر عن الهيئة القومية لضمان جودة التعليم والإعتماد.**

نسب توزيع المقررات وتناسقها مع المعايير المرجعية القياسية لقطاع الحاسبات

	Subject Area	NARS %	ISI Program %	Number of Courses	Number of Credit Hours
A	Humanities, Ethical and Social Sciences (Univ. Req.)	8-10	9.09	5 (Compulsory) 1 (Elective)	10 (Compulsory) 2 (Elective)
B	Mathematics and Basic Sciences	16-18	15.9	7 (Compulsory)	21 (Compulsory)
C	Basic Computing Sciences (institution req.)	26-28	29.55	13 (Compulsory)	39 (Compulsory)
D	Applied Computing Sciences (specialization)	28-30	29.64	12 (Compulsory) 1 (Elective)	36 (Compulsory) 3 (Elective)
E	Training	3-5	2.27	1 (Compulsory)	3 (Compulsory)
F	Projects	3-5	4.55	2 (Compulsory)	6 (Compulsory)
	Subtotal	84-96	91	40 (Compulsory) 2 (Elective)	115 (Compulsory) 5 (Elective)
G	Optional (Institution character-identifying subjects)	16-4	9	4 (Elective)	12 (Elective)
	Total	100	100		132

G. Humanities, Ethical and Social Sciences (Univ. Req.) 12 C.H 9.09%

• أولًا: متطلبات إجبارية (10 ساعة معتمدة) **Compulsory (10 Credit Hours)**

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
ENG101	English -1 لغة انجليزية - 1	-	2	2	-	-
ENG102	English - 2 لغة انجليزية - 2	ENG101 English - 1	2	2	-	-
PSC110	Human Rights حقوق الانسان	-	2	2	-	-
CPS121	Communication Skills and Report Writing مهارات التواصل و كتابة التقارير	ENG101 English - 1	2	-	4	-
CSE211	Ethics and Social Impacts of Computing الأخلاق والآثار الاجتماعية للحوسبة	-	2	2	-	-

• ثانياً: متطلبات إختيارية (2 ساعة معتمدة) **Electives (2 Credit Hours)**

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
PSY101	Psychology علم النفس	-	2	2	-	-
SOC101	Sociology علم الإجتماع	-	2	2	-	-
MGT103	Introduction to Management مبادئ إدارة	-	2	2	-	-
SCT101	Scientific Thinking التفكير العلمى	-	2	2	-	-
EPS101	Entrepreneurship ريادة الأعمال	-	2	2	-	-

H. Mathematics and Basic Sciences 21 C.H 15.9%

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
BMT111	Calculus تفاضل و تكامل	-	3	2	2	-
BMT121	Differential Equations معادلات تفاضليه	BMT111 Calculus	3	2	2	-
BMT212	Linear Algebra الجبر الخطي	BMT121 Differential Equations	3	2	2	-
BMT112	Discrete Mathematics تراكيب محددة	-	3	2	2	-
BPH111	Physics فيزياء	-	3	2	-	2
BMT122	Probability and Statistics إحصاء و إحتمالات	BMT111 Calculus	3	2	2	-
BMT211	Advanced Probability and Statistics إحصاء و احتمالات متقدمه	BMT122 Probability and Statistics	3	2	2	-

I. Basic Computing Sciences (institution req.) 39 C.H 29.55%

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
CSC111	Computing Fundamentals اساسيات الحاسب	-	3	2	1	2
CSC121	Structured Programming برمجه هيكلية	CSC111 Computing Fundamentals	3	2	1	2
CSC211	Object Oriented Programming برمجه شيئية	CSC121 Structured Programming	3	2	1	2
CSC221	Advanced Programming برمجة متقدمة	CSC211 Object Oriented Programming	3	2	1	2

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
CSC222	Data Structures هياكل البيانات	CSC211 Object Oriented Programming	3	2	1	2
CSC311	Operating Systems - 1 نظم تشغيل - 1	CSC211 Object Oriented Programming	3	2	-	2
CSC223	Software Engineering - 1 هندسة البرمجيات - 1	CSC211 Object Oriented Programming	3	2	-	2
CSC122	Logic Design التصميم المنطقي	BMT112 Discrete Mathematics	3	2	-	2
ISI222	Database Systems -1 نظم قواعد البيانات - 1	CSC121 Structured Programming	3	2	-	2
ISI311	Modeling and Simulation النمذجة والمحاكاة	BMT122 Probability and Statistics	3	2	-	2
CSS221	Computer Networks شبكات الحاسب	CSC111 Computing Fundamentals and BMT112 Discrete Mathematics	3	2	-	2
CSC312	Analysis and Design of Algorithms تحليل وتصميم الخوارزميات	CSC222 Data Structures	3	2	-	2
ISI211	Introduction to Information Systems مقدمة في نظم المعلومات	CSC111 Computing Fundamentals	3	2	-	2

J. Applied Computing Sciences (specialisation) + Institution Character-Identifying subjects 51 C.H 38.64%

Compulsory (39 Credit Hours) أولاً: متطلبات إجبارية (39 ساعة معتمدة)

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
ISI321	Database Systems - 2 نظم قواعد البيانات - 2	ISI222 Database Systems - 1	3	2	-	2
ISI312	Systems Analysis and Design - 1 تحليل وتصميم النظم - 1	ISI211 Introduction to Information Systems	3	2	-	2
CSA311	Artificial Intelligence Fundamentals أساسيات الذكاء الإصطناعي	CSC211 Object Oriented Programming	3	2	-	2
ISI421	Decision Support Systems نظم دعم القرار	ISI321 Database Systems-2	3	2	-	2
ISI422	Business Intelligence ذكاء الأعمال	ISI412 Data Mining and Data Warehousing	3	2	-	2
ISI322	Information Storage and Retrieval تخزين وأسترجاع البيانات	ISI222 Database Systems -1	3	2	-	2
ISI411	Management Information Systems نظم المعلومات الإدارية	ISI222 Database Systems - 1	3	2	-	2
ISI323	Geographical Information Systems نظم المعلومات الجغرافية	ISI222 Database Systems - 1	3	2	-	2
ISI412	Data Mining and Data Warehousing التنقيب في البيانات وتخزينها	ISI321 Database Systems -2	3	2	-	2
CSS312	Computers and Information Security أمن الحاسبات والمعلومات	CSS221 Computer Networks	3	2	-	2
CSC326	Web Applications Development تطوير تطبيقات الإنترنت	CSC211 Object Oriented Programming	3	-	-	6

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
	(Lab Course مقرر معمل)	and ISI222 Database Systems - 1				
ISI324	Software Project Management إدارة مشروعات البرمجيات	CSC223 Software Engineering - 1	3	2	-	2
ISI413	Enterprise Resource Planning تخطيط موارد المؤسسة	ISI222 Database Systems - 1	3	2	-	2

ثانياً: متطلبات إختيارية (12 ساعة معتمدة) Electives (12 Credit Hours)

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
ISI428	E-business and Digital Firms الكترونيات الأعمال و الشركات الرقمية	ISI312 Systems Analysis and Design - 1	3	2	-	2
ISI429	Systems Analysis and Design - 2 تحليل وتصميم النظم - 2	ISI312 Systems Analysis and Design - 1	3	2	-	2
ISI416	Health Care Information Systems نظم المعلومات الصحية	ISI222 Database Systems - 1	3	2	-	2
ISI325	Operations Research بحوث العمليات	BMT212 Linear Algebra	3	2	-	2
ISI326	Expert Systems نظم الخبرة	ISI222 Database Systems - 1	3	2	-	2
CSC426	Mobile Applications Development تطوير تطبيقات المحمول (Lab Course مقرر معمل)	CSC211 Object Oriented Programming	3	-	-	6
ISI423	Selected Topics in Information Systems - 1 موضوعات مختارة في نظم المعلومات - 1	ISI211 Introduction to Information Systems	3	2	-	2
ISI424	Selected Topics in Information Systems - 2	ISI211	3	2	-	2

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
	موضوعات مختارة في نظم المعلومات - 2	Introduction to Information Systems				
ISI415	Multimedia Database قواعد بيانات الوسائط المتعددة	ISI321 Database Systems -1	3	2	-	2
CSC411	Cloud Computing الحوسبة السحابية	CSC223 Software Engineering - 1 And CSS221 Computer Networks	3	2	-	2
ISD418	Social Media Mining التنقيب بوسائل التواصل الإجتماعي	ISI412 Data Mining and Data Warehousing	3	2	-	2
ISD411	Big Data Analytics تحليلات البيانات الضخمة	ISI412 Data Mining and Data Warehousing	3	2	-	2
ISI427	Knowledge Base Systems نظم المعرفة	ISI222 Database Systems - 1	3	2	-	2
ISI414	Human-Computer Interface Development تطوير واجهة المستخدم	CSC326 Web Applications Development	3	2	-	2

E. Training 3 E.H 2.27%

Course Code كود المقرر	Course Name اسم المقرر	Earned Hours	Contact Hours ساعات الإتصال		
			Lecture	Tutorial	Practical
TR333	Summer Training التدريب الصيفي	3	لايقل عن 12 ساعة أسبوعيا لمدة ست أسابيع		

- يتم تسجيل الطالب في مقرر التدريب الصيفي بشرط إجتيازة 70 ساعة معتمدة على الأقل.

F. Projects 6 C.H 4.55%

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
PR498	Project - 1 مشروع - 1	CSC223 Software Engineering -1	3	1	-	4
PR499	Project - 2 مشروع - 2	PR498 Project -1	3	1	-	4

مقترح الخطة الدراسية لبرنامج نظم المعلومات

الفصل الدراس ي الأول

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
ENG101	English - 1 لغة انجليزية - 1	2	-
BMT111	Calculus تفاضل وتكامل	3	-
BMT112	Discrete Mathematics تراكيب محددة	3	-
BPH111	Physics الفيزياء	3	-
CSC111	Computing Fundamentals اساسيات الحوسبة	3	-
PSC110	Human Rights حقوق الإنسان	2	-
Total **المجمـوع**		**16**	

الفصل الدراس ي الثاني

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
ENG102	English - 2 لغة انجليزية - 2	2	ENG101 English -1
BMT121	Differential Equations معادلات تفاضليه	3	BMT111 Calculus
CSC121	Structured Programming برمجه هيكليه	3	CSC111 Computing Fundamentals
CPS121	Communication Skills and Report Writing مهارات التواصل وكتابة التقارير	2	ENG101 English -1
BMT122	Probability and Statistics إحصاء وإحتمالات	3	BMT111 Calculus
CSC122	Logic Design التصميم المنطقي	3	BMT112 Discrete Mathematics
Total **المجمـوع**		**16**	

الفصل الدراس ي الثالث

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
CPS211	Ethics and Social Impacts of Computing الأخلاق والآثار الاجتماعية للحوسبة	2	-
CSC211	Object Oriented Programming برمجه شيئية	3	CSC121 Structured Programming
ISI211	Introduction to Information Systems مقدمة في نظم المعلومات	3	CSC111 Computing Fundamentals
BMT212	Linear Algebra الجبر الخطي	3	BMT121 Differential Equations
-	مقرر إختياري من (علوم إنسانية وأخلاقية واجتماعيه)	2	-
BMT211	Advanced Probability and Statistics إحصاء وإحتمالات متقدمه	3	BMT122 Probability and Statistics
Total **المجمـوع**		**16**	

الفصل الدراس ي الرابع

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
CSC221	Advanced Programming برمجة متقدمة	3	CSC211 Object Oriented Programming
CSC222	Data Structures هياكل البيانات	3	CSC211 Object Oriented Programming
CSC223	Software Engineering - 1 هندسة البرمجيات - 1	3	CSC211 Object Oriented Programming
CSS221	Computer Networks شبكات الحاسب	3	CSC111 Computing Fundamentals and BMT112 Discrete Mathematics
ISI222	Database Systems - 1 نظم قواعد البيانات - 1	3	CSC121 Structured Programming
Total **المجمـوع**		**15**	

الفصل الدراس ي الخامس

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
ISI311	Modeling and Simulation النمذجة والمحاكاه	3	BMT122 Probability and Statistics
CSA311	Artificial Intelligence Fundamentals أساسيات لذكاء الإصطناعي	3	CSC211 Object Oriented Programming
CSS312	Computers and Information Security أمن الحاسبات والمعلومات	3	CSS221 Computer Networks
ISI312	Systems Analysis and Design - 1 تحليل وتصميم النظم - 1	3	ISI211 Introduction to Information Systems
CSC312	Analysis and Design of Algorithms تحليل وتصميم الخوارزميات	3	CSC222 Data Structures
CSC311	Operating systems - 1 نظم التشغيل - 1	3	CSC211 Object Oriented Programming
Total **المجمـوع**		**18**	

الفصل الدراس ي السادس

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
CSC326	Web Applications Development تطوير تطبيقات الإنترنت	3	CSC211 Object Oriented Programming and ISI222 Database Systems - 1
ISI321	Database Systems - 2 نظم قواعد البيانات - 2	3	ISI222 Database Systems - 1
ISI322	Information Storage and Retrieval تخزين واسترجاع البيانات	3	ISI222 Database Systems -1
ISI323	Geographical Information Systems نظم المعلومات الجغرافية	3	ISI222 Database Systems - 1
ISI324	Software Project Management إدارة مشروعات البرمجيات	3	CSC223 Software Engineering - 1

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
-	Information Systems Program Elective - 1 اختياري برنامج نظم المعلومات - 1	3	*The prerequisite of the selected elective course
Total **المجمـوع**		**18**	

التدريب الصيفي

Course Code رقم المقرر	Course Name اسم المقرر	Earned Hours الساعات المكتسبة
TR333	Summer Training التدريب الصيفي	3

الفصل الدراس ي السابع

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
PR498	Project - 1 مشروع - 1	3	CSC223 Software Engineering -1
ISI411	Management Information Systems نظم المعلومات الإدارية	3	ISI222 Database Systems - 1
ISI412	Data Mining and Data Warehousing التنقيب عن البيانات وتخزينها	3	ISI321 Database Systems -2
ISI413	Enterprise Resource Planning تخطيط موارد المؤسسة	3	ISI222 Database Systems - 1
-	Information Systems Program Elective - 2 اختياري برنامج نظم المعلومات - 2	3	*The prerequisite of the selected elective course
Total **المجمـوع**		**15**	

الفصل الدراس ي الثامن

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
PR499	Project - 2 مشروع - 2	3	PR498 Project -1
ISI421	Decision Support Systems نظم دعم القرار	3	ISI321 Database Systems -2
ISI422	Business Intelligence ذكاء الأعمال	3	ISI412 Data Mining and Data Warehousing
-	Information Systems Program Elective - 3 اختياري برنامج نظم المعلومات - 3	3	*The prerequisite of the selected elective course
-	Information Systems Program Elective - 4 اختياري برنامج نظم المعلومات - 4	3	*The prerequisite of the selected elective course
Total **المجمـوع**		**15**	

Data Science Program - برنامج علوم البيانات

رسالة البرنامج

يلتزم برنامج علوم البيانات كلية الحاسبات وتكنولوجيا جامعة المستقبل بتقديم تعليم متميز يلبي احتياجات الخريجين لتطوير خبراتهم وقدراتهم على اكتساب المهارات العملية والمهنية والأخلاقيات في المجالات الحديثة والمتقدمة لعلوم البيانات وتحليلها بما يؤهلهم لتنفيذ الحلول والإستشارات لجهات الأعمال المختلفة في السوق المحلي والإقليمي والدولي، واجراء البحوث العلمية المبتكرة التي تساهم في التنمية المستدامة وخدمة المجتمع.

المهارات الخاصة بالبرنامج

بالإضافة الي المهارات العامة السالف ذكرها يكتسب خريج برنامج علوم البيانات المهارات الاتيه:

- استخدام الأدوات والتقنيات المناسبة لتجميع البيانات الأولية بأنواعها المختلفة وإعدادها للمعالجة أو التحليل النوعي.
- تطبيق التقنيات الإحصائية والتحليلية لتحويل هذه الكميات الكبيرة من البيانات الأولية وغير المهيكلة إلى معلومات ذات قيمة.
- تقديم البيانات وعرضها بصريًا لتوضيح الاستراتيجيات أو القرارات المبنية على نتائج تحليل تلك البيانات.
- استخلاص الأفكار والرؤى من البيانات المحللة باستخدام تقنيات الذكاء الاصطناعي وخوارزميات تعلم الآلة.
- وضع استراتيجيات تطوير للأعمال انطلاقًا من نتائج التحليل والدراسة وتطبيق النماذج وتقنيات الذكاء الاصطناعي وتعلم الآلة.
- بناء النماذج الكفء لوضع توقعات صحيحة أو قريبة من الصحة في مختلف المجالات مثل استطلاعات الرأي والانتخابات وحجوزات السفر واستكشاف حالات الغش، والدراسات العلمية ،والاجتماعية، وغيرها.

***يتم تحديث المهارات وفق مستجدات وإستقراء سوق العمل و NARS الصادر عن الهيئة القومية لضمان جودة التعليم والإعتماد.**

نسب توزيع مقررات البرنامج وتناسقها مع المعايير المرجعية القياسية لقطاع الحاسبات

	Subject Area	NARS %	ISD Program %	Number of Courses	Number of Credit Hours
A	Humanities, Ethical and Social Sciences (Univ. Req.)	8-10	9.09	5 (Compulsory) 1 (Elective)	10 (Compulsory) 2 (Elective)
B	Mathematics and Basic Sciences	16-18	15.9	7 (Compulsory)	21 (Compulsory)
C	Basic Computing Sciences (institution req.)	26-28	29.55	13 (Compulsory)	39 (Compulsory)
D	Applied Computing Sciences (specialization)	28-30	29.64	12 (Compulsory) 1 (Elective)	36 (Compulsory) 3 (Elective)
E	Training	3-5	2.27	1 (Compulsory)	3 (Compulsory)
F	Projects	3-5	4.55	2 (Compulsory)	6 (Compulsory)
	Subtotal	84-96	91	40 (Compulsory) 2 (Elective)	115 (Compulsory) 5 (Elective)
G	Optional (Institution character-identifying subjects)	16-4	9	4 (Elective)	12 (Elective)
	Total	100	100		132

A. Humanities, Ethical and Social Sciences (Univ. Req.) 12 C.H 9.09%

- أولًا: متطلبات إجبارية (10 ساعة معتمدة) **Compulsory (10 Credit Hours)**

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
ENG101	English - 1 لغة انجليزية - 1	-	2	2	-	-
ENG102	English - 2 لغة انجليزية - 2	ENG101 English - 1	2	2	-	-
PSC110	Human Rights حقوق الانسان	-	2	2	-	-
CPS121	Communication Skills and Report Writing مهارات التواصل و كتابة التقارير	ENG101 English - 1	2	-	4	-
CSE211	Ethics and Social Impacts of Computing الأخلاق والآثار الاجتماعية للحوسبة	-	2	2	-	-

- ثانياً: متطلبات إختيارية (2 ساعة معتمدة) **Electives (2 Credit Hours)**

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
PSY101	Psychology علم النفس	-	2	2	-	-
SOC101	Sociology علم الإجتماع	-	2	2	-	-
MGT103	Introduction to Management مبادئ إدارة	-	2	2	-	-
SCT101	Scientific Thinking التفكير العلمى	-	2	2	-	-
EPS101	Entrepreneurship ريادة الأعمال	-	2	2	-	-

B. Mathematics and Basic Sciences 21 C.H 15.9%

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
BMT111	Calculus تفاضل و تكامل	-	3	2	2	-
BMT121	Differential Equations معادلات تفاضليه	BMT111 Calculus	3	2	2	-
BMT212	Linear Algebra الجبر الخطي	BMT121 Differential Equations	3	2	2	-
BMT112	Discrete Mathematics تراكيب محددة	-	3	2	2	-
BPH111	Physics فيزياء	-	3	2	-	2
BMT122	Probability and Statistics إحصاء و إحتمالات	BMT111 Calculus	3	2	2	-
BMT211	Advanced Probability and Statistics إحصاء و احتمالات متقدمه	BMT122 Probability and Statistics	3	2	2	-

C. Basic Computing Sciences (institution req.) 39 C.H 29.55%

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
CSC111	Computing Fundamentals اساسيات الحاسب	-	3	2	1	2
CSC121	Structured Programming برمجه هيكلية	CSC111 Computing Fundamentals	3	2	1	2
CSC211	Object Oriented Programming برمجه شيئية	CSC121 Structured Programming	3	2	1	2
CSC221	Advanced Programming	CSC211	3	2	1	2

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
	برمجة متقدمة	Object Oriented Programming				
CSC222	Data Structures هياكل البيانات	CSC211 Object Oriented Programming	3	2	1	2
CSC311	Operating Systems - 1 نظم تشغيل - 1	CSC211 Object Oriented Programming	3	2	-	2
CSC223	Software Engineering - 1 هندسة البرمجيات - 1	CSC211 Object Oriented Programming	3	2	-	2
CSC122	Logic Design التصميم المنطقي	BMT112 Discrete Mathematics	3	2	-	2
ISI222	Database Systems - 1 نظم قواعد البيانات - 1	CSC121 Structured Programming	3	2	-	2
ISI311	Modeling and Simulation النمذجة والمحاكاة	BMT122 Probability and Statistics	3	2	-	2
CSS221	Computer Networks شبكات الحاسب	CSC111 Computing Fundamentals and BMT112 Discrete Mathematics	3	2	-	2
CSC312	Analysis and Design of Algorithms تحليل وتصميم الخوارزميات	CSC222 Data Structures	3	2	-	2
ISD211	Fundamentals of Data Science أساسيات علوم البيانات	CSC111 Computing Fundamentals	3	2	-	2

D. Applied Computing Sciences (specialisation) + Institution Character-Identifying subjects 51 C.H 38.64%

Compulsory (39 Credit Hours) أولًا: متطلبات إجبارية (39 ساعة معتمدة)

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
ISD322	Theory and Practice of Data Analysis نظرية وممارسة تحليل البيانات	CSA311 Artificial Intelligence Fundamentals	3	2	-	2
ISD311	Data wrangling and Knowledge Discovery معالجة البيانات واكتشاف المعرفة	ISI222 Database Systems - 1	3	2	-	2
CSA311	Artificial Intelligence Fundamentals أساسيات الذكاء الإصطناعي	CSC211 Object Oriented Programming	3	2	-	2
CSA411	Neural Networks and Deep Learning الشبكات العصبية والتعلم العميق	BMT121 Differential Equations and CSA311 Artificial Intelligence Fundamentals	3	2	-	2
ISD321	Computational Statistics and Data Analysis الإحصاء الحسابي وتحليل البيانات	BMT211 Advanced Probability and Statistics	3	2	-	2
ISD411	Big Data Analytics تحليلات البيانات الضخمة	ISD311 Data wrangling and Knowledge Discovery and ISD 211 Fundamentals of Data Science	3	2	-	2
ISD421	Data Visualization & Exploration تصور البيانات واستكشافها	ISD322 Theory and Practice of Data Analysis	3	2	-	2

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
ISD412	Applied Regression Methods تطبيقات الطرق المرجعية	BMT212 Linear Algebra and BMT211 Advanced Probability and Statistics	3	2	-	2
CSA321	Machine Learning تعلم الآلة	BMT122 Probability and Statistics and CSA311 Artificial Intelligence Fundamentals	3	2	-	2
ISD422	Applied Multivariate Analysis التحليل التطبيقي متعدد المتغيرات	ISD322 Theory and Practice of Data Analysis	3	2	-	2
CSS312	Computers and Information Security أمن الحاسبات والمعلومات	CSS221 Computer Networks	3	2	-	2
CSC411	Cloud Computing الحوسبة السحابية	CSC223 Software Engineering - 1 and CSS221 Computer Networks	3	2	-	2
CSC326	Web Applications Development تطوير تطبيقات الإنترنت (مقرر معمل Lab Course)	CSC211 Object Oriented Programming and ISI222 Database Systems - 1	3	-	-	6

ثانياً: متطلبات إختيارية (12 ساعة معتمدة) Electives (12 Credit Hours)

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
ISD413	GIS and Spatial Data Mining نظم المعلومات الجغرافية وتنقيب البيانات المكانية	ISD322 Theory and Practice of Data Analysis and ISD311 Data wrangling and Knowledge Discovery	3	2	-	2
ISD414	Optimization Algorithms خوارزميات التحسين	CSA321 Machine Learning	3	2	-	2
ISD415	Introduction to Block Chain & Distributed Ledgers مقدمة في تقنية سلسة الكتل و الحسابات الموزّعة	CSS312 Computers and Information Security	3	2	-	2
ISI326	Human-Computer Interface Development تطوير واجهة المستخدم	CSC211 Object Oriented Programming	3	2	-	2
ISD423	Intrusion Detection and Prevention أكتشاف والوقاية من التسلل والأختراقات	CSS312 Computers and Information Security	3	2	-	2
ISD424	Distributed Data Analysis تحليل البيانات الموزعة	ISI222 Database Systems - 1	3	2	-	2
ISD418	Social Media Mining التنقيب بوسائل التواصل الاجتماعي	ISD311 Data wrangling and Knowledge Discovery	3	2	-	2
ISD425	Data Compression Techniques تقنيات ضغط البيانات	CSC221 Advanced Programming and BMT211 Advanced Probability and Statistics	3	2	-	2

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
ISD416	Computational Linguistics اللغويات الحاسوبية	BMT212 Linear Algebra and BMT211 Advanced Probability and Statistics	3	2	-	2
ISD426	Selected Topics in Data Science - 1 موضوعات مختارة في علوم البيانات - 1	ISD211 Fundamentals of Data Science أساسيات علوم البيانات	3	2	-	2
ISD427	Selected topics in Data Science - 2 موضوعات مختارة في علوم البيانات - 2	ISD211 Fundamentals of Data Science أساسيات علوم البيانات	3	2	-	2

E. Training 3 E.H 2.27%

Course Code كود المقرر	Course Name اسم المقرر	Earned Hours	Contact Hours ساعات الإتصال		
			Lecture	Tutorial	Practical
TR333	Summer Training التدريب الصيفي	3	لايقل عن 12 ساعة أسبوعيا لمدة ست أسابيع		

- يتم تسجيل الطالب في مقرر التدريب الصيفي بشرط إجتيازة 70 ساعة معتمدة على الأقل.

F. Projects 6 C.H 4.55%

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
PR498	Project - 1 مشروع - 1	CSC223 Software Engineering -1	3	1	-	4
PR499	Project - 2 مشروع - 2	PR498	3	1	-	4

مقترح الخطة الدراسية لبرنامج علوم البيانات

الفصل الدراس ي الأول

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
ENG101	English - 1 لغة انجليزية - 1	2	-
BMT111	Calculus تفاضل وتكامل	3	-
BMT112	Discrete Mathematics تراكيب محددة	3	-
BPH111	Physics الفيزياء	3	-
CSC111	Computing Fundamentals اساسيات الحوسبة	3	-
PSC110	Human Rights حقوق الإنسان	2	-
Total **المجمـوع**		**16**	

الفصل الدراس ي الثاني

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
ENG102	English-2 لغة انجليزية -2	2	ENG101 English -1
BMT121	Differential Equations معادلات تفاضليه	3	BMT111 Calculus
CSC121	Structured Programming برمجه هيكليه	3	CSC111 Computing Fundamentals
CPS121	Communication Skills and Report Writing مهارات التواصل وكتابة التقارير	2	ENG101 English -1
BMT122	Probability and Statistics إحصاء وإحتمالات	3	BMT111 Calculus
CSC122	Logic Design التصميم المنطقي	3	BMT112 Discrete Mathematics
Total **المجمـوع**		**16**	

الفصل الدراس ي الثالث

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
CPS211	Ethics and Social Impacts of Computing الأخلاق والآثار الاجتماعية للحوسبة	2	-
CSC211	Object Oriented Programming برمجة شيئية	3	CSC121 Structured Programming
ISD211	Fundamentals of Data Science أساسيات علوم البيانات	3	CSC111 Computing Fundamentals
BMT212	Linear Algebra الجبر الخطي	3	BMT121 Differential Equations
-	مقرر إختياري من (علوم إنسانية وأخلاقيه وإجتماعية)	2	-
BMT211	Advanced Probability and Statistics إحصاء وإحتمالات متقدمه	3	BMT122 Probability and Statistics
Total **المجمـوع**		**16**	

الفصل الدراس ي الرابع

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
CSC221	Advanced Programming برمجة متقدمة	3	CSC211 Object Oriented Programming
CSC222	Data Structures هياكل البيانات	3	CSC211 Object Oriented Programming
CSC223	Software Engineering - 1 هندسة البرمجيات - 1	3	CSC211 Object Oriented Programming

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
CSS221	Computer Networks شبكات الحاسب	3	CSC111 Computing Fundamentals and BMT112 Discrete Mathematics
ISI222	Database Systems - 1 نظم قواعد البيانات - 1	3	CSC121 Structured Programming
Total المجموع		**15**	

الفصل الدراسي الخامس

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
ISI311	Modeling and Simulation النمذجة والمحاكاه	3	BMT122 Probability and Statistics
CSA311	Artificial Intelligence Fundamentals أساسيات لذكاء الإصطناعي	3	CSC211 Object Oriented Programming
CSS312	Computers and Information Security أمن الحاسبات والمعلومات	3	CSS221 Computer Networks
CSC311	Operating Systems - 1 نظم التشغيل - 1	3	CSC211 Object Oriented Programming
CSC312	Analysis and Design of Algorithms تحليل وتصميم الخوارزميات	3	CSC222 Data Structures
ISD311	Data Wrangling and Knowledge Discovery معالجة البيانات واكتشاف المعرفه	3	ISI222 Database Systems - 1
Total المجموع		**18**	

الفصل الدراس ي السادس

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
ISD321	Computational Statistics and Data Analysis الإحصاء الحسابي وتحليل البيانات	3	BMT211 Advanced Probability and Statistics
ISD322	Theory and Practice of Data Analysis نظرية وممارسة تحليل البيانات	3	CSA311 Artificial Intelligence Fundamentals
CSA321	Machine Learning تعلم الآلة	3	BMT122 Probability and Statistics and CSA311 Artificial Intelligence Fundamentals
CSC326	Web Applications Development تطوير تطبيقات الإنترنت	3	CSC211 Object Oriented Programming and ISI222 Database Systems - 1
-	Data Science Program Elective - 1 إختياري برنامج علوم البيانات - 1	3	*The prerequisite of the selected elective course
-	Data Science Program Elective - 2 إختياري برنامج علوم البيانات - 2	3	*The prerequisite of the selected elective course
Total المجموع		**18**	

التدريب الصيفي

Course Code رقم المقرر	Course Name اسم المقرر	Earned Hours الساعات المكتسبة
TR333	Summer Training التدريب الصيفي	3

الفصل الدراس ي السابع

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
CSA411	Neural Networks and Deep Learning الشبكات العصبية والتعلم العميق	3	BMT121 Differential Equations and CSA311 Artificial Intelligence Fundamentals
ISD411	Big Data Analytics تحليلات البيانات الضخمة	3	ISD311 Data wrangling and Knowledge Discovery And ISD 211 Fundamentals of Data Science
ISD412	Applied Regression Methods تطبيقات الطرق المرجعية	3	BMT212 Linear Algebra And BMT211 Advanced Probability and Statistics
CSC411	Cloud Computing الحوسبة السحابية	3	CSC223 Software Engineering - 1 And CSS221 Computer Networks
PR498	Project - 1 مشروع - 1	3	CSC223 Software Engineering -1
Total المجمـوع		**15**	

الفصل الدراس ي الثامن

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
ISD421	Data Visualization & Exploration تصور البيانات واستكشافها	3	ISD322 Theory and Practice of Data Analysis
ISD422	Applied Multivariate Analysis التحليل التطبيقي متعدد المتغيرات	3	ISD322 Theory and Practice of Data Analysis
-	Data Science Program Elective - 3 إختياري برنامج علوم البيانات - 3	3	*The prerequisite of the selected elective course

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
-	Data Science Program Elective - 4 إختياري برنامج علوم البيانات - 4	3	*The prerequisite of the selected elective course
PR499	Project - 2 مشروع - 2	3	PR498 Project - 1
Total **المجمـوع**		**15**	

برنامج تكنولوجيا الوسائط الرقمية

Digital Media Technology Program

رسالة البرنامج

يلتزم برنامج تكنولوجيا الوسائط الرقمية بكلية الحاسبات وتكنولوجيا المعلومات - جامعة المستقبل – بإعداد وتأهيل خريج متميز في التصميم الرقمي وصناعة الوسائط المتعددة باستخدام مهارات الاتصال والتفكير النقدي وحل المشكلات قادر على مواكبة التطور المستمر والمنافسة محليا وإقليميا ودوليا في سوق العمل، وإنتاج أبحاث علمية مبتكرة تخدم المجتمع وتظهر المشاركة الفعالة في تحقيق أهداف التنمية المستدامة مع الالتزام بالسلوك المهني والمعايير الأخلاقية.

المهارات الخاصة بالبرنامج

بالإضافة الي المهارات العامة السالف ذكرها يكتسب خريج برنامج تكنولوجيا الوسائط الرقمية المهارات الاتيه:

- فهم الرياضيات والإحصاءات الأساسية المطلوبة لحل المشكلات في تقنية الوسائط الرقمية. - التعرف على المفهوم الأساس ي للغات البرمجة عالية المستوى.
- شرح مبادئ وتقنيات المجالات المختلفة في مجال تكنولوجيا الوسائط الرقمية. - تحديد أدوار العوامل البشرية في تصميم أنظمة الوسائط الرقمية.
- تحديد الأدوات والتقنيات لتصميم التطبيقات وتطويرها.
- تحديد التحديات المختلفة الكامنة في صيانة أنظمة تكنولوجيا الوسائط الرقمية وتطورها. - تحليل مشاكل علوم الحاسوب المختلفة وتحديد الأهداف والمتطلبات.
- اختيار المنهجيات والأساليب المناسبة لحل مشكلة معينة وتحديد حدودها وقيودها وأخطاءها. - تصنيف البيانات والنتائج، والأساليب، والتقنيات، والخوارزميات.
- تحليل وصف المشكلات المختلفة ووضع مواصفات المتطلبات لها.
- تطبيق المعلومات الفعالة للحصول على تقنيات تخزين المعلومات واسترجاعها وإدارتها في صناعة الوسائط المختلفة والنصوص، والصور، والصوت، والفيديو.
- استخدم مبادئ تفاعل الإنسان مع الحاسب في بناء وتقييم واجهات المستخدم لنطاقات واسعة من تطبيقات تكنولوجيا الوسائط الرقمية.
- تطبيق المهارات الشخصية المختلفة عن طريق الوسائل الشفوية والكتابية والعروض التقديمية والمرئية بطريقة احترافية في مشروعات الوسائط المتعددة.
- التعرف على الأدوار المختلفة للعمل الجماعي في تطوير مشروعات تكنولوجيا الوسائط الرقمية. - تطبيق المسائل المهنية والأخلاقية في مشروعات تكنولوجيا الوسائط الرقمية.
- استخدم أدوات تكنولوجيا الوسائط الرقمية بشكل فعال لإدارة المشروعات البرمجية.
- تطبيق مهارات مختلفة مثل التعلم الذاتي وإدارة الوقت في التعلم طويل المدى.

***يتم تحديث المهارات وفق مستجدات وإستقراء سوق العمل و NARS الصادر عن الهيئة القومية لضمان جودة التعليم والإعتماد.**

نسب توزيع مقررات البرنامج وتناسقها مع المعايير المرجعية القياسية لقطاع الحاسبات

	Subject Area	NARS %	DMT Program %	Number of Courses	Number of Credit Hours
A	Humanities, Ethical and Social Sciences (Univ. Req.)	8-10	9.09	5 (Compulsory) 1 (Elective)	10 (Compulsory) 2 (Elective)
B	Mathematics and Basic Sciences	16-18	15.9	7 (Compulsory)	21 (Compulsory)
C	Basic Computing Sciences (institution req.)	26-28	29.55	13 (Compulsory)	39 (Compulsory)
D	Applied Computing Sciences (specialization)	28-30	29.64	12 (Compulsory) 1 (Elective)	36 (Compulsory) 3 (Elective)
E	Training	3-5	2.27	1 (Compulsory)	3 (Compulsory)
F	Projects	3-5	4.55	2 (Compulsory)	6 (Compulsory)
	Subtotal	84-96	91	40 (Compulsory) 2 (Elective)	115 (Compulsory) 5 (Elective)
G	Optional (Institution character-identifying subjects)	16-4	9	4 (Elective)	12 (Elective)
	Total	100	100		132

A. Humanities, Ethical and Social Sciences (Univ. Req.) 12 C.H 9.09%

- أولًا: متطلبات إجبارية (10 ساعة معتمدة) **Compulsory (10 Credit Hours)**

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
ENG101	English-1 لغة انجليزية -1	-	2	2	-	-
ENG102	English-2 لغة انجليزية -2	ENG101 English-1	2	2	-	-
PSC110	Human Rights حقوق الانسان	-	2	2	-	-
CPS121	Communication Skills and Report Writing مهارات التواصل و كتابة التقارير	ENG101 English-1	2	-	4	-
CSE211	Ethics and Social Impacts of Computing الأخلاق والآثار الاجتماعية للحوسبة	-	2	2	-	-

- ثانياً: متطلبات إختيارية (2 ساعة معتمدة) **Electives (2 Credit Hours)**

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
PSY101	Psychology علم النفس	-	2	2	-	-
SOC101	Sociology علم الإجتماع	-	2	2	-	-
MGT103	Introduction to Management مبادئ إدارة	-	2	2	-	-
SCT101	Scientific Thinking التفكير العلمى	-	2	2	-	-
EPS101	Entrepreneurship ريادة الأعمال	-	2	2	-	-

B. Mathematics and Basic Sciences 21 C.H 15.9%

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
BMT111	Calculus تفاضل و تكامل	-	3	2	2	-
BMT121	Differential Equations معادلات تفاضليه	BMT111 Calculus	3	2	2	-
BMT212	Linear Algebra الجبر الخطي	BMT121 Differential Equations	3	2	2	-
BMT112	Discrete Mathematics تراكيب محددة	-	3	2	2	-
BPH111	Physics فيزياء	-	3	2	-	2
BMT122	Probability and Statistics إحصاء و إحتمالات	BMT111 Calculus	3	2	2	-
BMT211	Advanced Probability and Statistics إحصاء و احتمالات متقدمه	BMT122 Probability and Statistics	3	2	2	-

C. Basic Computing Sciences (institution req.) 39 C.H 29.55%

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
CSC111	Computing Fundamentals اساسيات الحاسب	-	3	2	1	2
CSC121	Structured Programming برمجه هيكلية	CSC111 Computing Fundamentals	3	2	1	2
CSC211	Object Oriented Programming برمجه شيئية	CSC121 Structured Programming	3	2	1	2
CSC221	Advanced Programming برمجة متقدمة	CSC211 Object Oriented Programming	3	2	1	2

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
CSC222	Data Structures هياكل البيانات	CSC211 Object Oriented Programming	3	2	1	2
CSC311	Operating Systems - 1 نظم تشغيل - 1	CSC211 Object Oriented Programming	3	2	-	2
CSC223	Software Engineering - 1 هندسة البرمجيات - 1	CSC211 Object Oriented Programming	3	2	-	2
CSC122	Logic Design التصميم المنطقي	BMT112 Discrete Mathematics	3	2	-	2
ISI222	Database Systems -1 نظم قواعد البيانات	CSC121 Structured Programming	3	2	-	2
ISI311	Modeling and Simulation النمذجة والمحاكاة	BMT122 Probability and Statistics	3	2	-	2
CSS221	Computer Networks شبكات الحاسب	CSC111 Computing Fundamental sand BMT112 Discrete Mathematics	3	2	-	2
CSC312	Analysis and Design of Algorithms تحليل وتصميم الخوارزميات	CSC222 Data Structures	3	2	-	2
DMT212	Introduction to Media Technology مقدمة في تكنولوجيا الوسائط (مقرر معمل Lab Course)	CSC111 Computing Fundamentals	3	-	-	6

D. Applied Computing Sciences (specialisation) + Institution Character-Identifying subjects 51 C.H 38.64%

Compulsory (39 Credit Hours) أولًا: متطلبات إجبارية (39 ساعة معتمدة)

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
DMT412	Game Development تطوير الألعاب	CSA311 Artificial Intelligence Fundamentals and DMT324 Introduction to 2D animation	3	2	-	2
DMT322	Video and Audio Technology تكنولوجيا الصوت والصورة	DMT311 Digital Signals Processing	3	2	-	2
DMT421	Virtual Reality الواقع الافتراضي (مقرر معمل Lab Course)	DMT412 Game Development	3	-	-	6
DMT422	Augmented Reality الواقع المعزز	CSC211 Object Oriented Programming	3	2	-	2
DMT311	Digital Signals Processing معالجة الإشارات الرقمية	BMT121 Differential Equations	3	2	-	2
DMT323	Computer Graphics الرسم بالحاسب	BMT212 Linear Algebra	3	2	-	2
CSS312	Computers and Information Security أمن الحاسبات والمعلومات	CSS221 Computer Networks	3	2	-	2
DMT325	Character Design for Film and Games تصميم الشخصيات للأفلام و الألعاب	DMT 212 Introduction to Media Technology	3	2	-	2
CSC326	Web Application Development تطوير تطبيقات الانترنت (مقرر معمل Lab Course)	CSC211 Object Oriented Programming and ISI222 Database Systems -1	3	-	-	6
DMT324	Introduction to 2D animation مقدمة في الرسوم المتحركة ثنائية الابعاد (مقرر معمل Lab Course)	DM 212 Introduction to Media Technology	3	-	-	6

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
DMT411	Introduction to 3D animation مقدمة في الرسوم المتحركة ثلاثية الابعاد	DMT324 Introduction to 2D animation	3	2	-	2
DMT326	User Experience Design تصميم تجربة المستخدم	CSC223 Software Engineering - 1	3	2	-	2
CSA311	Artificial Intelligence Fundamentals أساسيات الذكاء الإصطناعي	CSC211 Object Oriented Programming	3	2	-	2

ثانياً: متطلبات إختيارية (12 ساعة معتمدة) **Electives (12 Credit Hours)**

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
DMT423	Distributed Multimedia Applications تطبيقات الوسائط المتعددة الموزعة	CSS221 Computer Networks	3	2	-	2
DMT414	Visual Effects مؤثرات بصرية	DMT324 Introduction to 2D animation	3	2	-	2
CSS313	Advanced Computer Networks شبكات الحاسب المتقدمة	CSS221 Computer Networks	3	2	-	2
ISI324	Software Projects Management إدارة مشروعات البرمجيات	CSC223 Software Engineering - 1	3	2	-	2
DMT425	Speech Processing معالجة الكلام	DMT311 Digital Signals Processing	3	2	-	2
DMT416	Video Processing معالجة الفيديو	DMT311 Digital Signals Processing	3	2	-	2
CSC426	Mobile Applications Development تطوير تطبيقات المحمول (مقرر معمل Lab Course)	CSC211 Object Oriented Programming	3	-	-	6
CSA313	Image Processing معالجة الصور	BMT212 Linear Algebra and CSC121	3	2	-	2

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
		Structured Programming				
DMT426	Selected Topics in Digital Media Technology -1 موضوعات مختاره في تكنولوجيا الوسائط الرقمية-1	DMT212 Introduction to Media Technology	3	2	-	2
DMT427	Selected Topics in Digital Media Technology -2 موضوعات مختاره في تكنولوجيا الوسائط الرقمية-2	DMT212 Introduction to Media Technology	3	2	-	2

E. Training 3 E.H 2.27%

Course Code كود المقرر	Course Name اسم المقرر	Earned Hours	Contact Hours ساعات الإتصال		
			Lecture	Tutorial	Practical
TR333	Summ التدريب الصيفي	3	لايقل عن 12 ساعة أسبوعيا لمدة ست أسابيع		

- يتم تسجيل الطالب في مقرر التدريب الصيفي بشرط إجتيازة 70 ساعة معتمدة على الأقل.

F. Projects 6 C.H 4.55%

Course Code كود المقرر	Course Name اسم المقرر	Prerequisite المتطلب السابق	C.H.	Contact Hours ساعات الإتصال		
				Lecture	Tutorial	Practical
PR498	Project-1 مشروع-1	CSC223 Software Engineering -1	3	1	-	4
PR499	Project-2 مشروع-2	PR498	3	1	-	4

الفصل الدراس ي الأول

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
ENG101	English-1 لغة انجليزية -1	2	-
BMT111	Calculus تفاضل وتكامل	3	-
BMT112	Discrete Mathematics تراكيب محددة	3	-
BPH111	Physics الفيزياء	3	-
CSC111	Computing Fundamentals اساسيات الحوسبة	3	-
PSC110	Human Rights حقوق الإنسان	2	-
Total المجمـوع		**16**	

الفصل الدراس ي الثاني

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
ENG102	English-2 لغة انجليزية -2	2	ENG101 English -1
BMT121	Differential Equations معادلات تفاضليه	3	BMT111 Calculus
CSC121	Structured Programming برمجه هيكليه	3	CSC111 Computing Fundamentals
CPS121	Communication Skills and Report Writing مهارات التواصل وكتابة التقارير	2	ENG101 English -1
BMT122	Probability and Statistics إحصاء وإحتمالات	3	BMT111 Calculus
CSC122	Logic Design التصميم المنطقي	**3**	BMT112 Discrete Mathematics
Total المجمـوع		**16**	

الفصل الدراس ي الثالث

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
CPS211	Ethics and Social Impacts of Computing الأخلاق والآثار الاجتماعية للحوسبة	2	-
CSC211	Object Oriented Programming برمجه شيئية	3	CSC121 Structured Programming
DMT212	Introduction to Media Technology مقدمة في تكنولوجيا الوسائط	3	CSC111 Computing Fundamentals
BMT212	Linear Algebra الجبر الخطي	3	BMT121 Differential Equations
-	مقرر إختياري من (علوم إنسانية وأخلاقية واجتماعيه)	2	-
BMT211	Advanced Probability and Statistics إحصاء وإحتمالات متقدمه	3	BMT122 Probability and Statistics
Total **المجمـوع**		**16**	

الفصل الدراس ي الرابع

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
CSC221	Advanced Programming برمجة متقدمة	3	CSC211 Object Oriented Programming
CSC222	Data Structures هياكل البيانات	3	CSC211 Object Oriented Programming
CSC223	Software Engineering - 1 هندسة البرمجيات - 1	3	CSC211 Object Oriented Programming
CSS221	Computer Networks شبكات الحاسب	3	CSC111 Computing Fundamentals and BMT112 Discrete Mathematics

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
ISI222	Database Systems - 1 نظم قواعد البيانات	3	CSC121 Structured Programming
Total المجمـوع		**15**	

الفصل الدراس ي الخامس

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
ISI311	Modeling and Simulation النمذجة والمحاكاة	3	BMT122 Probability and Statistics
CSA311	Artificial Intelligence Fundamentals أساسيات الذكاء الإصطناعي	3	CSC211 Object Oriented Programming
CSS312	Computers and Information Security أمن الحاسبات والمعلومات	3	CSS221 Computer Networks
DMT311	Digital Signal Processing معالجة الإشارات الرقمية	3	BMT121 Differential Equations
CSC312	Analysis and Design of Algorithms تحليل وتصميم الخوارزميات	3	CSC222 Data Structures
CSC311	Operating systems -1 نظم التشغيل -1	3	CSC211 Object Oriented Programming
Total المجمـوع		**18**	

الفصل الدراس ي السادس

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
CSC326	Web Applications Development تطوير تطبيقات الإنترنت	3	CSC211 Object Oriented Programming and ISI222 Database Systems -1
DMT322	Video and Audio Technology تكنولوجيا الصوت والصورة	3	DMT311 Digital Signals Processing
DMT323	Computer Graphics الرسم بالحاسب	3	BMT212 Linear Algebra
DMT324	Introduction to 2D animation مقدمة في الرسوم المتحركة ثنائية الابعاد	3	DM 212 Introduction to Media Technology
DMT325	Character Design for Film and Games تصميم الشخصيات للأفلام و الألعاب	3	DMT 212 Introduction to Media Technology
DMT326	User Experience Design تصميم تجربة المستخدم	3	CSC223 Software Engineering - 1
Total المجمـوع		**18**	

التدريب الصيفي

Course Code رقم المقرر	Course Name اسم المقرر	Earned Hours الساعات المكتسبة
TR333	Summer Training التدريب الصيفي	3

الفصل الدراس ي السابع

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
PR498	Project-1 مشروع -1	3	CSC223 Software Engineering - 1
DMT411	Introduction to 3D animation مقدمة في الرسوم المتحركة ثلاثية الابعاد	3	DMT324 Introduction to 2D animation
DMT412	Game Development تطوير الألعاب	3	CSA311 Artificial Intelligence Fundamentals And DMT324 Introduction to 2D animation
-	Digital Media Technology Program Elective-1 إختياري برنامج تكنولوجيا الوسائط الرقمية 1	3	*The prerequisite of the selected elective course
-	Digital Media Technology Program Elective-2 إختياري برنامج تكنولوجيا الوسائط الرقمية 2	3	*The prerequisite of the selected elective course
Total **المجموع**		**15**	

الفصل الدراس ي الثامن

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
PR499	Project-2 مشروع -2	3	PR498 Project-1
DMT421	Virtual Reality الواقع الافتراض ي	3	DMT412 Game Development
DMT422	Augmented Reality الواقع المعزز	3	CSC211 Object Oriented Programming

Course Code رقم المقرر	Course Name اسم المقرر	Credit Hours الساعات المعتمدة	Prerequisite المتطلب السابق
-	Digital Media Technology Program Elective3 إختياري برنامج تكنولوجيا الوسائط الرقمية 3	3	*The prerequisite of the selected elective course
-	Digital Media Technology Program Elective4 إختياري برنامج تكنولوجيا الوسائط الرقمية 4	3	*The prerequisite of the selected elective course
Total **المجمـوع**		**15**	

الباب الثالث: توصيف المقررات

Humanities, ethical and Social Sciences

English - 1 (ENG101) **لغة انجليزية - 1**

The course provides an academic reading and writing proficiencies that enabling students to build on the reading and writing skills they have developed throughout the years and to progress toward an advanced level of literacy. Topics covered in this course are reading comprehension, writing fundamentals, essay writing, creative writing, vocabulary building, literary analysis and the core transferable skills in critical thinking, reading and writing that students will use in their university courses, in addition, analysis of a variety of challenging readings.

English - 2 (ENG102) **لغة انجليزية - 2**

The course provides a wide variety of reading and writing skills that develop the student's language practice, in addition to longer reading materials that enhance understanding of various academic topics that are considered the main focus of this course.Topics covered in this course are topical and functional lexis provide the reference by which language is introduced and recycled within clear natural contexts, writing is explored through a range of texts, by understanding genre specific conventions, and developing confidence by planning and discussions and by applying both process and product approaches.

Human Rights (PSC110) **حقوق الإنسان**

The course provides the conceptual foundation of human rights, the development of human rights framework and multi- disciplinary character of the field as an area of the study. Topics covered in this course are the universal declaration on human rights, the international covenant on economic, social, and cultural rights and all related agreements. It will also touch on the state of human rights in egypt and the arab world.

Communication Skills and Report Writing (CPS121) **مهارات التواصل وكتابة التقارير**

The course provides students with abilities to effectively communicate and convey information through well-structured reports and developing essential skills necessary for producing concise, clear, and professional reports that cater to different audiences and purposes. Topics covered in this course are the planning, drafting, editing, and presentation phases, organizing information logically, employing appropriate language and tone. Also, covering both written and oral communications and ensuring students can articulate their ideas effectively through written reports as well as presentations.

Ethics and Social Impacts of Computing (CSE211) **الأخلاق والآثارالاجتماعية للحوسبة**

The course provides critical examination of ethical problems associated with computer science and engineering, legal and quasi-legal (i.e., policy and regulative) issues. Topics covered in the course are intellectual property and copyright, process of ethical decision-making, privacy and confidentiality, computer crime, professional codes and responsibilities, software piracy and impact of computers on society.

Entrepreneurship (EPS101) **ريادة الأعمال**

The course provides the meaning of entrepreneurship, its importance to the Entrepreneur and to the economy. Topics covered in the course are knowing the importance of an entrepreneurial idea, the means and sources of generating ideas, identify the characteristics, aptitudes, qualifications of an effective entrepreneur, identify different areas and fields that entrepreneur can choose among them to start his/her project, prepare and develop a feasibility study for the project, formulate a strategic/business plan for the project, manage the project professionally, identify the organizations that can support and incubate the entrepreneurial projects.

Psychology (PSY101) **علم نفس**

The course provides different types of psychological behavior and factors affecting it. The course also, deals with the different problems and behavior of youth. Topics covered in the course are introduction to psychology, biological bases on behavior, sensation and perception, state of consciousness, memory, learning, social psychology, personality and personality disorders.

Sociology (SOC101) **علم اجتماع**

The course provides the students with an understanding of the basic principles of sociology as an academic discipline and provide an analytical perspective of society and everyday life through sociological theories. The course helps the students to analyze the ways in which people interact and function in groups. Topics covered in the course are culture, values, norms, social stratification, social problems, race and ethnicity, conformity, deviance, urban living, social change, and social movements.

Introduction to Management (MGT103) **مبادئ الإدارة**

The course provides the overall functions of management and organizational structure and dynamics. Topics covered in the course are definition and evolution of management, the functions of management, managerial roles and skills, organizational structure, leadership and motivation, discussion on decision-making process, including the identification of problems, generating alternatives, evaluating options, and making choices, communication in management through

highlighting the importance of effective communication methods and channels. and exploring the barriers of teaching specific skills in cross cultural competence, planning, quality initiatives, human resource management, leadership, teamwork, and decision making.

Scientific Thinking (SCT 101) **التفكيرالعلمي**

The course provides students with a basic understanding of scientific thinking. Students will be exposed to concepts, terminology, principles, and theories that comprise a course in thinking scientifically. Topics covered in the course are to synthesize the broad range of knowledge about thinking scientifically, to emphasize research methodology, to encourage critical thinking, and to convey a scientific as well as systematic approach to think over a concept.

Mathematics and Basic Sciences

Calculus (BMT101) **تفاضل وتكامل**

The course provides the fundamental principles and techniques of calculus, which is a branch of mathematics that deals with rates of change and accumulation. Topics covered in this course are functions, limits, continuity, differentiation and rules of differentiation, chain rule, implicit differentiation, differentiation of parametric functions, transcendental functions and differentiation. Also, the course covers trigonometric and inverse trigonometric functions, exponential and logarithmic functions, hyperbolic and inverse hyperbolic functions, indeterminate forms and l'hopital's rule, application of derivatives, taylor and maclaurin expansion and extrema of a function, integration concepts, transcendental functions and techniques of integration like integration by substitutions, integration by parts and integration by partial fractions.

Differential Equations (BMT121) **معادلات تفاضلية**

The course provides students with a basic understanding of the fundamental theorem of calculus, integration by trigonometric substitutions, integration by reduction , applications of definite integral: area, volume, arclengh of a curve. polynomial and series. Topics covered in the course are testing for convergence and divergence of a series, ordinary differential equations like equations of first order (separable, homogenous, exact, linear and Bernoulli), orthogonal trajectories, equations reducible to first order, high order linear equations, the variation of parameters , operation method, Euler's equation, and system of linear differential equations.

Linear Algebra (BMT212) **الجبرالخطي**

The course provides the basic understanding, definitions and properties of determinants and matrices, algebra of matrices, reduced matrix, rank of a matrix, solution of linear systems using inverse matrix, and Cramer's Rule. Topics covered in this course are homogeneous and non-homogeneous systems, square and rectangular systems, idea of a Linear transformation, matrix of linear transformation, solution of nonlinear algebraic equations by fixed point method, secant method, solution of linear algebraic systems by Iterative methods, jacobi method, Seidel Method and solution of nonlinear algebraic systems by Gauss elimination, Gauss elimination with pivot, LU decomposition method, Gauss - Jordan method Eigenvalues and Eigenvectors. Also, the course introduces orthogonality, projections, least squares approximations, orthonormal bases, and linear regression.

Discrete Mathematics (BMT112) **تراكيب محددة**

The course provides the concept of Preliminaries. Topics covered in this course are sets, binary relations, functions, summations, matrices; Logic: language of logic, quantifiers, advanced induction; Formal Logic: propositional calculus, Boolean algebra; Algorithms: pseudo-code language, recursive algorithms; Graph Theory: graphs, digraphs, paths, cycles, adjacency matrices, connectivity, trees, graph algorithms; Enumeration: permutations, combinations, inclusion/exclusion, Binomial Theorem, combinatorial identities; Group Theory: groups, subgroups, cosets, Lagrange's Theorem, quotient groups; Difference Equations/Recurrence Relations: basic types of recurrences/difference equations and solution methods.

Physics (BPH111) **الفيزياء**

The course provides students with a foundational understanding of fundamental aspects of physics, focusing on key topics related to electrical and magnetic fields.Topics covered in this course are electrical field, electrical potential energy, electrical current, Coulomb's law, Kirchhoff's Law, and Ampere's Law. Additionally, the course explores quantization, black body radiation, Bohr Model of the Hydrogen atom, and the photoelectric effect, all of which are essential examples of magnetic fields and forces. Hands-on provisions will include practical experiments or simulations to reinforce the theoretical concepts and their real-world applications.

Advanced Physics (BPH211) **الفيزياء المتقدمة**

The course provides students with a comprehensive exploration of various topics in physics, focusing on oscillatory and wave motion, optics, and quantum physics. Topics covered in this course include oscillatory motion, wave motion, sound waves, Doppler effect, superposition and standing waves,

principles of ray optics, Huygens' Principle, reflection, and refraction. Wave optics topics include interference, diffraction patterns, and polarization. The course also introduces important principles such as Max-Planck's principle and provides an introduction to quantum physics, covering atomic physics, molecules, and solids. Hands-on provisions include laboratory experiments and/or demonstrations to enhance understanding and practical application of the theoretical concepts.

Probability and Statistics (BMT122) **الإحصاء والإحتمالات**

The course provides students with a fundamental understanding of probability theory and its applications in statistics. Topics covered in this course include basic probability concepts, probability theorems, conditional probability, independent and dependent events, the total probability rule, Bayes' theorem, discrete and continuous probability distributions, random variables, statistical principles, statistical analysis of data and various statistical measures. Hands-on provisions include practical exercises and/or data analysis projects to apply the learned concepts in real-world scenarios.

Advanced Probability and Statistics (BMT211) **الإحصاء والاحتمالات المتقدمة**

The course provides statistical inference and its practical applications through the construction of mathematical models and solving statistical problems. Topics covered in the the course are understanding of sampling distribution (distribution of mean), central limit theorem, concept of estimation theory, point estimation, some properties (maximum likelihood method , Moment method), interval estimation (population mean and variance, two population mean and variance), concept of testing hypothesis (population mean and variance, two population mean and variance), chi-square test, introduction to correlation and regression. Hands-on provisions include data analysis projects and statistical software usage to reinforce the theoretical concepts and enhance practical skills in statistical inference.

Number Theory (BMT213) **نظرية الأعداد**

The course provides an introduction to number theory. Topics covered in this course are the integers and natural numbers, divisibility, congruence arithmetic, the Euclidean algorithm, greatest common divisors, Bézout coefficients, the fundamental theorem of arithmetic (unique factorization), solving linear congruences, the Chinese remainder theorem, Euler's totient function $\varphi(n)$, arithmetic functions, quadratic reciprocity, arithmetic properties of Fibonacci numbers, Pell's equation, continued fractions, prime numbers, Fermat primes, Mersenne primes, prime number theorem, Bertrand's postulate, Goldbach conjecture, Riemann's hypothesis, partitions, Pascal's triangle, sums of squares, representations, Pythagorean triples, Lagrange's four square theorem, rational approximations, linear diophantine equations, primality testing, factorization techniques, primitive roots and order.

Basic Computing Sciences

Computing Fundamentals (CSC 111) **أساسيات الحاسب**

The course provides students with the basic knowledge of the computing discipline and the necessary skills how to use computers. Topics covered in this course are introduction to computers, data representation and number systems, computer hardware, computer software, computer networks, the internet and its emerging applications, cybersecurity, computing professional ethics and social impact, and problem solving methods (algorithmic method). Hands-on is provided through a series of exercises, assignments and projects.

Structured Programming (CSC121) **برمجه هيكليه**

The course provides students with the basic knowledge and skills of structured programming. Topics covered in the course are fundamentals of problem solving and program development, structured programming basics, programming of decisions, programming of repetitions, modular programming using functions, arrays, pointers, dynamic memory allocation, and File I/O data streams. Hands-on is provided through a series of exercises, assignments, and projects.

Object Oriented Programming (CSC211) **برمجه شيئيه**

The course provides the advanced concepts of solving and programming computing problems on the computer. Topics covered in this course are methods, arrays, concepetual basis of OOP, classes, methods, inheritance, abstractation and polymorphism. Hands-on experience is provided through a series of exercises, assignments and projects.

Advanced Programming (CSC221) **برمجه متقدمة**

The course provides advanced approaches for developing and creating rich desktop applications to program, build, implement, and integrate complex components and objects in practical real-world applications. The course topics include modelling data using programming language type systems, event and exception programming, thread programming, persistence and distributed programming. Hands on experience is provided through a series of exercises and projects.

Data Structures (CSC222) **هياكل البيانات**

The course provides basic principles of data structures in computing, including design, verification, and application in terms of both structural elements and algorithmic strategies. Topics covered in this course are basic concepts of the data structures, arrays, static implementation of stacks and queues, linked lists, dynamic implementation of stacks and queues, trees, binary trees, binary search, graphs, sorting

algorithms, searching algorithms and hashing. Hands-on experience is provided through a series of exercises, assignments, and projects.

Operating Systems - 1 (CSC311) **نظم التشغيل - 1**

This course provides an overview of operating system concepts, a comprehensive survey of operating system concepts, structures, and processes. Topics covered in this course are OS support for concurrency (threads) and synchronization, deadlocks, process scheduling, Memory management, multi-programming, swapping, virtual memory, paging , symbolic segmentation, multithread programming, inter-process communication, and distributed interactions. Hands-on experience is provided through a series of exercises, assignments, and projects.

Software Engineering - 1 (CSC223) **هندسة البرمجيات - 1**

The course provides software engineering discipline with emphasis on Software engineering ethics and Professional software development. Topics covered in this course are software process models including waterfall model, incremental model, reuse oriented model and The rational unified process, agile software development principals and other methods such as extreme programming and scrum, requirements engineering including functional, non functional requirements and requirement document development, system modelling including structural model, behavioural model and architecture design. Hands-on experience is provided through a series of exercises, assignments, and projects.

Logic Design (CSC122) **التصميم المنطقي**

The course provides the fundamentals of logic, including design, verification, application, programmable logic arrays and memories. In addition, the various logical structures and families are discussed in detail. Topics covered in this course are basic logic gates, boolean functions, simplification using boolean algebra, boolean function, simplification using K-Maps, combinational circuit design, adders, subtracts, multiplication, division circuits, multiplexer, de-multiplexer, encoders, and decoders. Hands-on experience is provided through a series of exercises, assignments, and projects.

Database Systems - 1 (ISI222) **نظم قواعد البيانات - 1**

The course provides students with a comprehensive understanding of the core concepts and principles of database management systems (DBMS).Topics covered in this course are understanding the role of databases in modern applications, types of databases (relational, NoSQL, etc.), relational model, structured query language (SQL), database design and normalization, overview of popular DBMSs

(e.g., MySQL, PostgreSQL, Microsoft SQL Server), data manipulation language (DML), data querying with SQL, data constraints and integrity, indexing and performance optimization, ACID properties of transactions, dealing with concurrent access and avoiding data inconsistencies, introduction to NoSQL databases and real-world applications and case studies. Hands-on experience is provided through a series of exercises, assignments, and projects.

Modeling and Simulation (ISI311) **النمذجة والمحاكاة**

This course provides the essential principles and methods of modelling and simulation of both discrete-event and continuous-time (dynamic) systems.Topics covered in this course are steps of simulation study with examples, statistical models (discrete and continuous time simulation models), simulation of queuing systems (single server or multi servers), probability distributions and inverse transform method to generate random variates, random-number generation, uniformity and independency statistical tests for the generated random numbers, inventory system, comparison and evaluation or alternative system designs and output analysis for a single model. Hhands-on experience is provided through a series of exercises, assignments, and projects.

Computer Networks (CSS221) **شبكات الحاسب**

ThE course provides the fundamentals of computer networking. Topics covered in this course are the Internet architecture and protocols, the fundamental concepts, and principles of OSI and TCP/IP layers, the Ethernet communication technology, switching techniques, network performance metrics, data link layer, network layer, routing protocols, IP networking and addressing, transport layer protocols (TCP and UDP), Application layer protocols (e.g., BGP, SSH, SMTP, DNS, HTTPS, among others), Wireless and mobile networks technologies, network traffic analysis, the client-server model. Hhands-on experience is provided through a series of exercises, assignments, and projects.

Analysis and Design of Algorithms (CSC312) **تحليل وتصميم الخوارزميات**

The course provides the fundemntals of algorithm design and analysis, the basics of analyzing algorithms for correctness , time and space bounds. Up-completeness theory, the notion of intractable problems, approximation measures, polynomial time approximation, and algorithm complexity . Topics covered in this course are analyzing and designing algorithms (sorting algorithms), complexity types and analysis, recurrence equations iteration methods , heap property function and operations, search algorithms, raph concepts and operations, general greedy method., branch & bound algorithms, dynamic programming and string and pattern matching. Hands-on experience is provided through a series of exercises, assignments, and projects.

Introduction to Information Systems (ISI211) **مقدمة في نظم المعلومات**

The course provides the fundamental concepts of information systems and their applications in real-life activities, basic components, functions and procedures of a typical computer-based information system. Topics covered in this course are foundation of information systems in business, the competence with information technology, e-commerce and e-business, decision support systems, security and ethical challenges, enterprise and global management of information technology and inter-organizational systems and virtual organizations. Hands-on experience is provided through a series of exercises, assignments, and projects.

Fundamental of data Science (ISD211) **أساسيات علوم البيانات**

The course provides the core principles, methodologies, and tools used in the field of data science with the fundamental concepts and techniques required to extract valuable insights from large and complex datasets, and to make data-driven decisions. Topics covered in this course are data manipulation and preprocessing, Exploratory Data Analysis (EDA) such as data visualization techniques, descriptive statistics, and statistical analysis. In addition to machine learning concepts, data mining and pattern recognition, data ethics and privacy and real-world data science case studies. Hands-on experience is provided through a series of exercises, assignments, and projects.

Cyber Security Fundamentals (CSS311) **أساسيات الأمن السيبراني**

The course provides general cybersecurity principles. Topics covered in this course are cyber space components, cyber security domains, cyber security frameworks, and security architecture, cyber security concepts, cyber security risk assessments and mitigation techniques, understanding cybersecurity offense and defense, the role of cybersecurity professionals, legal and ethical principles, potential threats and malware types, the corresponding security controls, and countermeasures; physical security, network attacks and defenses, operating system security, application security (web, apps, and databases) and wireless security. In addition, general concepts in information security and information assurance, technical, operational, and organizational issues of securing information systems, common attack and defense strategies for software, web applications, networks, operating systems, cryptographic systems and humans will be explored. Hands-on experience is provided through a series of exercises, assignments, and projects.

Introduction to Media Technology (DMT212) **مقدمة في تكنولوجيا الوسائط**

The course provides the student with a fundamental understanding of the media assets that comprise interactive media experience and how they are made. Topics covered in this course are background design, color theory, enabling students to create visually engaging and expressive animated scenes.

Additionally, visual storytelling techniques and workflows for creating media assets using industry standard software will be covered .

توصيف المقررات علوم الحاسب

Computer Architecture (CSC323) **عمارة الحاسبات**

The course provides fundamentals of current computer architecture and design, which is a rapidly evolving discipline, covering the major themes in computer hardware architecture and design, allowing students to understand and apply basic and advanced concepts in the field. Topics covered in this course are registers, counters, register transfer language, bus and memory transfer, arithmetic micro-operations, logic micro-operations, shift micro-operations, instruction codes, computer registers, computer instructions, timing cycle, instruction cycle, memory reference instructions, input-output and interrupt, design of basic computer, design of accumulator logic, control memory, address sequencing, microprogrammed control unit, memory hierarchy. Hands on experience is provided through a series of exercises, assignments, and projects.

Software Engineering - 2 (CSC313) **هندسة البرمجيات - 2**

The course provides comprehensive coverage of the fast-evolving discipline of software engineering that enables students to design, construct, test, and evaluate a software system in a variety of real-world domains. Topics covered in this course are software architectural design, object-oriented design, software testing, software verification and validation, software reuse and critical system development. Hands on experience is provided through a series of exercises, assignments, and projects.

Operating Systems - 2 (CSC322) **نظم التشغيل - 2**

The course provides the main advanced subject areas of the standard operating system including operating system kernel functions and architecture, services, and user interface. Topics covered in this course are file systems, mass storage structure, input/output systems, system protection and system security. Hands on experience is provided through a series of exercises, assignments, and projects.

Theory of Computations (CSC324) **نظرية الحسابات**

The course provides theoretical concepts of computing discipline, such as abstract models of computing machines, the grammar that those machines understand, and the language classes that correspond to those grammars.

Topics covered in this course are formal languages theory, regular expressions, finite automata, non-deterministic finite automata, transition graphs, context-free grammars, pushdown automata, computing paradigms, turing machines and post machines. Students will gain practical experience through a series of exercises, assignments, and a course project.

Computer Organization & Assembly Language (CSC325) **تنظيم الحاسبات ولغة التجميع**

The course provides the principles of computer organization in computing discipline, including design, verification, application, components of a typical computer, including their functions and inter-communications. Topics covered in this course are ALU design, organization of the processor data path and control, performance analysis, memory system hierarchy, pipelining, communication, concepts of assembly language and how to program and manipulate computer basic through a low-level assembly language. Hands-on experience is provided through a series of exercises, assignments, and projects.

Embedded Systems (CSC412) **النظم المدمجة**

The course provides advanced concepts and techniques of Embedded Systems software development, basic concepts of embedded systems, disciplined development, and organization aspects of embedded systems.Topics covered in this course are embedded system peripherals, sensors, actuators, and interfacing, installing embedded systems building blocks, memory architectures and embedded processors, interrupts and multitasking, model-based design, dynamic modeling, state machine concepts, troubleshooting, advanced microcontroller-based tools and real-world embedded systems applications and case studies. Hands-on experience is provided through a series of exercises, assignments, and projects.

Concepts of Programming Languages (CSC422) **مفاهية لغات البرمجة**

The course provides analytical comparison and evaluation of different programming languages and a comparative study of syntax, concepts of Identifiers, data types, sequence control, run-time resources, translators, subprograms and their implementation, concurrency, and programming paradigms. Topics covered in this course are syntax, semantics, language design and implementation, abstraction of programming languages, binding and scopes, expressions and assignment statements, exception handling, event handling and analytical comparison and evaluation of different programming languages. Hands-on experience is provided through a series of exercises, assignments, and projects.

Cloud Computing (CSC411) **الحوسبة السحابية**

The course provides hands-on comprehensive study of cloud concepts and capabilities across the various cloud service models including infrastructure as a service (IaaS), platform as a service (PaaS), software as a service (SaaS), and business process as a service (BPaaS). Topics covered in the course are SaaS, use of vendor-maintained applications and processes available on the cloud on a metered on-demand basis in multitenant environments, cloud security model, associated challenges, implementation of a system based on service- oriented architecture, performance, virtualization, cloud mobility, security, cost, usability, and utility of cloud computing solutions. Hands on experience is provided through a series of exercises, assignments, and projects.

Distributed Processing (CSC423) **المعالجة الموزعة**

The course provides advanced concepts and techniques of parallel and distributed systems that emphasizes network, internetworking and distributes operating systems. Topics covered are characterization of distributed systems, distributed systems models, remote invocation, modelling specifications, consistency, fault tolerance, deadlock and live lock and starvation, inter-process communication and algorithms that run in parallel. Hands on experience is provided through a series of exercises, assignments, and projects.

Mobile Applications Development (CSC426) **تطويرتطبيقات المحمول**

The course provides mobile application technologies enabling students to design and develop mobile application software while putting emphasis on strengths and limitations of mobile devices, terminologies, and concepts.Topics covered in this course are introduction to mobile applications development, XML and android studio, app fundamentals and android GUI, activity lifecycle, intents and fragments, UI events and event listeners, shared preferences, SQLite databases and files handling and content providers. Hands on experience is provided through a series of exercises, assignments, and projects.

Web Applications Development (CSC326) **تطويرتطبيقات الانترنت**

The course provides students with the practical skills to develop emerging Web applications using the state-of-the-art technologies and development tools. The course is a project-based in which the student develops a complete web application for a selected emerging business domain. The course covers the following main subject areas: Web application architecture, Web application frameworks, Design and development of front end, Design and development of back end, Managing data, Middleware, and Implementation of some semantics Web features.

Software Testing (CSC327) **اختبار البرمجيات**

The course provides students with the skills to select and apply a testing strategy and testing techniques that are appropriate to a particular software system or component, in addition the student will become a capable user of test tools by assessing the effectiveness of their testing activity and providing evidence to justify their evaluation. Topics covered in this course are basic software testing techniques like black-box testing, white box testing, performance testing, configuration testing, integration testing, compatibility testing, test automation, various measurements needed to assess the adequacy of test suites, test maturity model, defects lifecycle, the structure of a test plan, test management, defect management, testing tools, automation, documentation, different techniques of testing- such as risk based testing, structured testing and advanced analytical, and statistical quality management techniques. Hands on experience is provided through a series of exercises, assignments, and projects.

Software Design & Architecture (CSC415) **عمارة وتصميم البرمجيات**

The course provides the continuation of the study of design patterns, frameworks, and architectures to create reusable, flexible, and maintainable software applications, and systems, as well as the expression and documentation of software systems.Topics covered in this course are design patterns, survey of current middleware architectures, design of distributed systems using middleware, component-based design, measurement theory and appropriate use of metrics in design, designing for qualities such as performance, safety, security, reusability, reliability, etc., measuring internal qualities and complexity of software, evaluation and evolution of designs, basics of software evolution, reengineering, and reverse engineering. Hands on experience is provided through a series of exercises, assignments, and projects.

Selected Topics in Computer Science - 1 (CSC414) **موضوعات مختارة في علوم الحاسب - 1**

The course provides specific areas of interest and emerging topics within the field of Computer Science. The course focuses on in-depth discussions and practical applications of selected topics, providing students with specialized knowledge and skills to tackle complex Computer Applications challenges. The specific topics covered may vary depending on the instructor and current trends in Computer science.

Selected Topics in Computer Science - 2 (CSC424) **موضوعات مختارة في علوم الحاسب - 2**

This course provides specific areas of interest and emerging topics within the field of Computer Science. The course focuses on in-depth discussions and practical applications of selected topics, providing students with specialized knowledge and skills to tackle complex Computer Applications challenges. The specific topics covered may vary depending on the instructor and current trends in Computer science. Hands on experience is provided through a series of exercises, assignments, and projects.

توصيف المقررات الذكاء الاصطناعي

Reasoning and Agents (CSA322) **أستنتاج وتمثيل المعرفة**

The course provides the knoweldge related to the representation, reasoning and planning for solving real world inference. The course illustrates the importance of using a smart representation of knowledge such that it is conducive to efficient reasoning, and the need for exploiting task constraints for intelligent search and planning. The notion of representing action, space and time is formalized in the context of agents capable of sensing the environment and taking actions that affect the current state. There is also a strong emphasis on the ability to deal with uncertain data in real world scenarios and hence, the planning and reasoning methods are extended to include inference in probabilistic domains. The course includes the subject areas: Introduction to Intelligent agents, nature of agents, performance measures and environments, search based planning, planning as a search problem, smart searching using constraints and exploiting subproblems, informed search and Exploration for agents, logical representation and planning, scaling planning for complex tasks, acting in uncertain (real world) Environments. Hands on experience is provided through a series of exercises, assignments, and projects.

Neural Networks and Deep Learning (CSA411) **الشبكات العصبية والتعلم العميق**

The course provides the foundational concept of neural networks, how to build and train neural network, deep neural network architectures and learn how to make them better with different strategies. The course includes the subject areas: Introduction to Neural Networks, Neural Network Structures, Learning Process and Rosenblatt's Perceptron, Associative Memory, LMS Algorithm for Single Layer Network, Multilayer Perceptron And Back-Propagation Algorithm, Unsupervised learning Neural Nets, Adaptive Resonance Theory (ART), Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN). Hands-on is provided through a series of exercises, assignments, and projects.

Artificial Intelligence Fundamentals (CSA311) **أساسيات الذكاء الاصطناعي**

The course provides basic knowledge about the rapidly evolving subject area of Artificial Intelligence (AI), and its emerging applications. It covers a range of basic AI topics and techniques; concepts in Artificial Intelligence, Problem solving agents, Uninformed and Informed search techniques, Game playing, knowledge representation, reasoning and inference in Propositional and First Order logic, and basic concept of machine learning. The course provides students with practical hands-on experience through a set of exercises, assignments, and a course project using Python language

Processing Formal and Natural Languages (CSA324) **معالجة اللغات الرسمية والطبيعية**

The course provides how to design NLP applications that perform question-answering and sentiment analysis, create tools to translate languages and summarize text. The course includes the subject areas: Grammars and the Chomsky Hierarchy, Regular languages, Finite state automata (FSA), probabilistic FSAs, Context-free languages and Push-down automata, Ambiguity and solutions to the problem, Deterministic parsers, Chart parsers, Probabilistic context-free grammars, Modelling semantics, Context-sensitive languages, Turing machines and computability, Models of human language processing, Overview of language technology. Hands-on is provided through a series of exercises, assignments, and projects.

Machine Learning (CSA321) **تعلم الألة**

The course provides students an introduction to machine learning and statistical pattern recognition. The course includes the subject areas: supervised learning and unsupervised learning techniques, neural networks, support vector machines, clustering and dimensionality reduction methods, and introduction to reinforcement learning. The course will also discuss recent applications of machine learning, such as robotic control, data mining, autonomous navigation, bioinformatics, speech recognition, and text and web data processing. In this course primarily the Python programming language will be used based on the familiarity with linear algebra, probability theory, and programming in Python. The course provides students with practical hands-on experience through a set of exercises, assignments, and a course project.

Image Processing (CSA313) **معالجة الصور**

The course provides basic concepts and methodologies for the formation, representation, enhancement, and analysis of digital images. It also provides a foundation for developing applications and gives an introduction for further study in pattern recognition and computer vision. The course includes the subject areas: image processing fundamentals, image transforms, image enhancement, image restoration and reconstruction, feature extraction, colour image processing. the course offers hands-on experience in the design and implementation of image processing algorithms through a set of assignments and course project.

Reinforcement Learning (CSA424) **التعلم التعزيزي**

The course provides an introduction to agents learning techniques where an agent explicitly takes actions and interacts with the world. The course includes the subject areas: overview of reinforcement learning, Bandit problems and online learning, Markov decision processes, Returns and value functions, Solution methods: dynamic programming, Monte Carlo learning, Temporal difference learning, Eligibility traces, Value function (VF) approximation, Deep Learning, Convnets and back-propagation, Deep Q-learning: Double Q-Learning, replay memory, Policy gradients, Continuous actions, Variational autoencoders, multimodal stochastic policies, Imitation learning, Hierarchical RL and transfer learning , Learning to learn, one shot learning. Hands-on is provided through a series of exercises, assignments, and projects.

Intelligent Autonomous Robotics (CSA422) **الانسان الالي ذاتي التحكم الذكي**

The course provides design principles for intelligent autonomous systems. It applies the knowledge in practice through the construction of a simple autonomous robot. The course includes the subject areas: history of robotics, actuators, locomotion, manipulation, sensors and sensing for robotics, sensor based and odometry navigation, workspace decomposition and search algorithms on graphs, configuration space and configuration-space obstacles, state estimation and dead reckoning, Bayesian filtering: Kalman and particle filters, trajectory planning, sampling-based motion planning, PID control, control using fuzzy logic, hierarchical control, simultaneous localization and mapping, multi-robots and applications. Hands-on is provided through a series of exercises, assignments, and projects.

Artificial Intelligence Applications (CSA323) **تطبيقات الذكاء الإصطناعى**

The course provides an overview of AI and its role in business transformation. The purpose of this course is to investigate the many ways in which AI is being used in the society, and provide a strategic framework for how to bring AI to the center of digital transformation efforts. In terms of society

applications, the course considerm applications of AI in media, finance, healthcare, retail, and other industries. Finally, the course considers how AI can be used as a source of competitive advantage . Hands-on is provided through a series of exercises, assignments, and projects.

Soft Computing (CSA416) **الحوسبة المرنة**

The course provides the basic concepts of Soft Computing methodology. The course includes the subject areas: Genetic Algorithms, Chromosomes, Population, Fitness functions, Crossover, Mutation, Binary bit chromosomes, Floating point array chromosomes, Schema theory, Fuzzy logic, Fuzzy systems, Fuzzy operators, Fuzzy rule-based systems, Neural networks, Feed forward neural networks, Back propagation algorithm, Bias, Scaling, Proof of Delta rule, Performance issues, Hybrid systems, Feature selection, Training of NNs with Gas, Evolution of fuzzy rule based systems, Genetic programming, Immune systems, Evolution strategy. Hands-on is provided through a series of exercises, assignments, and projects.

Bioinformatics Systems (CSA416) **نظم المعلومات الحيوية**

The course provides a fundamental concepts and methods in bioinformatics, a field at the intersection of biology and computing. The course includes the subject areas: Biological background related to bioinformatics, the genome, protein and motif databases, DNA replication, motifs finding algorithms, local and global pairwise sequence alignment, scoring matrices, introduction to multiple sequence alignment, genome assembly algorithms, microarray gene expression databases applications on microarrays datasets, genome compression. Hands-on is provided through a series of exercises, assignments, and projects.

Data Analytics (CSA426) **تحليل البيانات**

The course provides an introduction to Data Analytics. The course includes the subject areas: design, build, verify, and test predictive data models to make data-driven decisions, concepts, methods, and applications of pattern discovery in data mining, concepts of cluster analysis, clustering methodologies, algorithms, and applications, partitioning methods, hierarchical methods, and density-based methods, steps of data preparation, analysis, learning and modeling, and identifying the predictive/descriptive model that produces the best results. Hands-on is provided through a series of exercises, assignments, and projects.

Computational Cognitive Science (CSA425) **الحسابيه العلوم الادراكية**

The course provides an introduction to computational theories of human cognition and Formal models from classic and contemporary artificial intelligence. The course includes the subject areas: Introduction to Logic Introduction, Foundations of Inductive Learning, Concept Learning and Categorization, Controlling Complexity and Occam's Razor, Reasoning about Natural Kinds, Causal Learning and Theorems, Structure and Formation of Intuitive theories of Physical, Biological and Social Systems, Acquisition of Natural Language (syntax and semantics), Behavior and Mental States of Other People, Formal modeling topics. Hands-on is provided through a series of exercises, assignments, and projects.

Computer Vision (CSA423) **رؤية الآلة**

The course provides an introduction to computer vision. The course includes the subject areas: fundamentals of image formation, Camera imaging geometry, Feature detection and matching, Multiview geometry including stereo, Motion estimation and tracking, classification, Methods for depth recovery from stereo, Camera. Calibration, Image stabilization, Automated alignment (e.g. panoramas), Tracking and action recognition. Hands-on is provided through a series of exercises, assignments and projects.

Selected Topics in Artificial Intelligence 1 & 2 (CSA415, CSA428) **موضوعات مختارة في الذكاء الإصطناعي**

The course provides specific areas of interest and emerging topics within the field of Artificial Intelligence. The course focuses on in-depth discussions and practical applications of selected topics, providing students with specialized knowledge and skills to tackle complex Artificial Intelligence Applications challenges. The specific topics covered may vary depending on the instructor and current trends Artificial Intelligence.

Artificial Intelligence for Cyber Security (CSA412) **الذكاء الاصطناعي في الأمن السيبراني**

The course provides the knowledge about the impact of artificial intelligence in enhancing systems security. The course includes the subject areas: introduction, using Artificial Intelligence tools to enhance security, machine learning applications in modern cyber security and threat detection, botnet detection, intrusion detection, deep packet inspection, fraud monitoring, malware detection, phishing detection, active authentication, cyber penetration testing techniques and tools, politics, Big data and decision-making process, Artificial Intelligence in politics, future of AI in advancing cyber security.

Pattern Recognition (CSA413) **التعرف على الأنماط**

The course provides theory and algorithms of pattern recognition techniques. The course explores the methods of pattern recognition that are used in many applications such as information retrieval, data mining, document image analysis and recognition, computational linguistics, forensics, biometrics and bioinformatics. The course includes the subject areas: Bayesian decision rule, Quality metrics in pattern recognition, Supervised learning with parametric distributions, Supervised learning with non-parametric distributions, classification, Linear discriminant functions, single-layer perceptron - Support vector machines (SVMs), Multi-layer perceptron, neural networks (NNs), Unsupervised learning, clustering methods. Hands-on is provided through a series of exercises, assignments and projects.

توصيف المقررات الدراسية الأمن السيبراني

Advanced Computer Networks (CSS313) **شبكات الحاسب المتقدمة**

The course provides the practical aspects of computer networks with emphasis on the Internet. Topics covered in this course are various aspects of computer networking, alternative link-layer, network-layer, and transport-layer technologies, LAN/WAN technologies, topologies, traffic analysis, congestion/flow control, routing, internetworking, multicast, security, Quality of Service (QoS), Integrated services (IntServ.), Differentiated services (DiffServ.). The course also covers wireless technologies and their applications in areas such as IoT and sensor networks, Multimedia on the Internet; and Real-Time interactive protocols (RTP, RTCP). The course addresses emerging networking technologies such as future routing protocols, IPv6 transition, and software-defined networking. Hands-on is provided through a series of exercises, assignments, and projects.

Computers and Information Security (CSS312) **أمن الحاسبات والمعلومات**

The course provides the basic principles of computer and information security. Topics covered in this course are cryptography: symmetric ciphers, asymmetric ciphers, MACs and hash functions, digital signatures, Cryptographic protocols: identification, authorization, authentication and key agreement, anonymity protocols, Fundamental concepts of access control models, threat modelling; hardware security, Usable security: basic human factors, warning design, phishing, device authentication, Security standards and protocols, virtual private networks, OS security: OS concepts, memory and file system, access control, file permissions, memory safety, stack-based buffer overflows, Malware: viruses, trojan horses, worms, rootkits, DNS attacks, Firewalls, Database security. Hands-on is provided through a series of exercises, assignments, and projects.

Cryptography and Network Security (CSS321) **التشفيروأمن الشبكات**

The course provides students with a practical and theoretical knowledge of cryptography and network security. Topics covered in this course are symmetric encryption algorithms and design principles, Public-key encryption algorithms, Message authentication codes and hash functions, Digital signatures and public-key certificates, Key management protocols, Network security including attacks, vulnerabilities and defense measures, secure network design, Network, and transport layers security, Advanced cryptography, access control, distributed authentication, TCP/IP security, firewalls, IPSec, Virtual Private Networks, intrusion detection systems, Network level attacks and various defense mechanisms, Wireless networking security and how to configure, attack, and defend wireless networks, Attack and defense on common network services such as DNS, HTTP, SQL, and FTP, DoS, Web security: WWW basics, client-side attacks, session hijacking XSS, CSRF, defenses, server-side attacks, and SQL injection. Hands-on is provided through a series of exercises, assignments, and projects.

Software Security (CSS322) **أمن البرمجيات**

The course explores the foundations of software security. Topics covered in this course are important software vulnerabilities and attacks such as buffer overflows, SQL injection, and session hijacking. The course explores defenses that prevent and/or mitigate these attacks, including advanced testing and analysis techniques. It explores techniques for analyzing software to determine whether it contains weaknesses that are vulnerable to exploitation. The course addresses the concept of debugging, and testing secure software to ensure its capability of mitigating vulnerabilities. It also explores reverse engineering of software to understand the design of an existing software component to determine its security and whether it could contain malware. Hands-on is provided through a series of exercises, assignments, and projects.

Security Threats and Risk Analysis (CSS323) **التهديدات الأمنية وتحليل المخاطر**

The course provides an evaluation of the systems, software tools, and techniques deployed to protect digital assets in companies, enterprises and governmental organizations. Topics covered in this course are theories of data security to analyze potential vulnerabilities in Internet and enterprise architecture and design, loss prevention strategies, how to determine the levels of protection and response to security incidents, designing a consistent and reasonable information security system with appropriate intrusion detection and reporting features, inspection and protection of information assets, detection of and reaction to threats to information assets, and examination of pre- and post- incident procedures,

in depth risk management models that exist for implementing a deeper risk management program in the organization. Hands-on is provided through a series of exercises, assignments, and projects.

Ethical Hacking and Penetration testing (CSS324) **القرصنة الأخلاقية واختبارالاختراق**

The course provides the theory and practical techniques of ethical hacking and penetration testing, which are essential elements in modern cybersecurity. topics covered in this course are hacking and ethical hacking concepts, using tools for scanning and vulnerability assessment, hacking wireless networks, hacking into computers and smartphones, ethical hacking, systems' scanning, vulnerability detection, tools for hacking, cracking encryption, flaws in websites and applications; attacking with frameworks, penetration testing, phishing, advantages and disadvantages of WEP, WPA, and WPA2, resources wireless hacking, malware based attacks, man-in-the-middle attack; vulnerability assessment and penetration testing of web servers and web application servers. Hands-on is provided through a series of exercises, assignments, and projects.

Digital Forensics (CSS411) **التحاليل الجنائية الرقمية**

The course provides a practical approach to learning forensic computing techniques using opensource and commercial forensic tools. Topics covered in this course are fundamental data structures applicable to digital forensics and how various tools can be exploited to analyze these structures in a variety of case types, the fundamentals of computer systems and computer forensics, file system forensic analysis, mobile device forensics, malware analysis and multimedia forensics, the required skills and expertise to conduct a digital forensics investigation successfully, how to enable students to practice on a variety of digital devices activities including searching and seizing, acquisition, chain-of-custody, analysis, preservation, and reporting of digital evidence. Hands-on is provided through a series of exercises, assignments, and projects.

Mobile and Wireless Security (CSS421) **أمن المحمول واللاسلكي**

The course provides a security solutions in wireless and mobile networks. Topics covered in this course are the basic security concepts and fundamentals knowledge necessary for understanding and evaluating security issues, challenges, and solutions, security issues and available solutions associated with off-the-shelf wireless and mobile technologies such as Bluetooth, WiMax, 4G, and 5G, WiFi standards and protocols such as IEEE 802.11, IEEE 802.15 and IEEE 802.20 and security standards such as WEP and WPA, security techniques and solutions used to protect applications downloaded by mobile terminals through mobile-cellular networks, security issues and solutions in emerging wireless and mobile technologies such as ad hoc and sensor networks, cellular 5G and IMS networks.

Cyber Security Incident Detection and Response (CSS422) **كشف حوادث الأمن السيبراني والاستجابة لها**

The course prepares students for incident detection and response within an organization and the application of incident detection and response practices to address a variety of cybersecurity incidents. Topics covered in this course are incident detection and response in organizations, understanding cybersecurity incidents, challenges of cybersecurity incidents, preparing for cybersecurity incidents, detection and classification of network events, identifying cybersecurity incidents, responding to cybersecurity incidents; cybersecurity incident analysis, mitigation, recovery, and reporting; incident recording and documentation, incident reporting; common security mitigation tools, and common cybersecurity incidents handling. Hands-on is provided through a series of exercises, assignments, and projects.

Internet of Things (IoT) (CSS325) **إنترنت الأشياء**

The course provides an exhaustive illustration of the enabling components of IoT systems. Topics covered in this course are wireless standards for short and long M2M communication (e.g. BLE, 6LoWPAN, LoRa, and NB-IoT) among sensor edge-devices, language sand boards for the development of prototypal WSAN, WPAN, and WLAN sensor networks (e.g. Arduino, STM32Nucleo, Raspberry PI,…), IoT network stacks and protocols, architectures and tools for data storage and processing, including sensor time-series database and fog/cloud approaches and services (e.g., Amazon AWS IoT and Thing Speak), techniques for the valorization of sensor data and knowledge extraction from large IoT deployment sand crowd-sensing environments through the application of machine-learning and data analytics techniques (e.g., time-series prediction, classification, clustering, anomaly detection, etc.). Hands-on is provided through a series of exercises, assignments, and projects.

Cyber-Physical Security (CSS327) **الأمن السيبراني المادي**

The course examines the security and privacy issues in the vast implementations of Cyber-physical systems (CPS). The course explores how existing security mechanisms can be applied to the CPS system and why such protections are not enough. Topics covered in the course are characteristics of CPS, security and privacy concerns, risk assessment for CPSs, attacks against CPSs; high-profile real-world attacks against CPSs, preventing attacks, detecting attacks, mitigating attacks, combined safety and security analysis of cyber-physical systems, policy and political aspects of CPS Security, industry practices and standards, security for specific CPS classes: industrial control systems, electric power grids, transportation systems and autonomous vehicles, robotics and advanced manufacturing,

medical devices and the Internet of Things. Hands-on is provided through a series of exercises, assignments, and projects.

Cloud Security (CSS423) **أمن الحوسبة السحابية**

The course describes the Cloud security architecture and explores the guiding security design principles, design patterns, industry standards, applied technologies and addressing regulatory compliance requirements critical to design, implement, deliver, and manage secure cloud-based services. Topics covered in this course are secure cloud architectural aspects with regards to identifying and mitigating risks, protection and isolation of physical and logical infrastructures including computing, network, and storage, how to apply comprehensive data protection at all OSI layers, end-to-end identity management and access control, monitoring and auditing processes and meeting compliance with industry and regulatory mandates, cloud computing security guidelines set forth by ISO, NIST, ENISA and Cloud Security Alliance (CSA). Hands-on is provided through a series of exercises, assignments, and projects.

Cybersecurity Law and Policy (CSS326) **قانون وسياسة الأمن السيبراني**

The course provides the basic principles of cybersecurity law and policy. Topics covered in this course are the technological overview of the Internet, cryptography, and computer hacking, the technological, legal, and policy dimensions around a wide range of current cybersecurity issues across the following categories: cybersecurity in the private sector, cybercrime and digital investigations, and the international dimensions of cybersecurity, the national and international legal frameworks that govern malicious and defensive actions in cyberspace including laws related to cybercrime, cyberespionage, and cyberwar, legal questions within the context of broader debates about issues such as the roles of governmental and non-governmental actors and the role of law in governing a constantly changing domain where many actors operate in secret. Hands-on is provided through a series of exercises, assignments, and projects.

Web Application Security and Testing (CSS412) **أمن تطبيقات الويب واختباره**

The course discusses web applications from an offensive security standpoint. Topics covered in this course are information gathering, vulnerability detection, infiltration, and privilege escalation, web application architecture, penetration testing a web application, and hardening a vulnerable application, reconnaissance, relevant tooling used in industry, server-side attacks such as SQL injection, local file inclusion, and server-side request forgery, client-side vulnerabilities such as cross site scripting, authentication, session management; access control and privilege escalation, common

web service vulnerabilities, patching and remediation, secure web coding best practice, vulnerability reporting, professional communication, and pen-testing in industry. Hands-on is provided through a series of exercises, assignments, and projects.

Advanced Penetration Testing (CSS413) **اختبارات الاختراق المتقدمة**

The course explores the defender side. Topics covered in this course are IDS/IPS overview and bypasses, antivirus bypasses, binary evasion and packing binaries, attacking from the web: Cross-site scripting (XSS) attacks, SQL injection attack, remote file inclusion, amd local file inclusion, exploiting the network: client side (browser) exploitation, windows privilege escalation, Linux privilege escalation, re-purposing administrative tools, post exploitation: Windows persistence techniques, enabling RDP/VNC, Linux persistence techniques, backdooring SSH logins, post exploitation reconnaissance, data exfiltration techniques, attacks against network access control (NAC) and virtual local area network (VLAN) manipulation, network device exploitation, breaking out of Linux and Windows restricted environments, IPv6, Linux privilege escalation and exploit-writing, and testing cryptographic implementation. Hands-on is provided through a series of exercises, assignments, and projects.

Malicious Software (CSS427) **البرمجيات الخبيثة**

The course aims at providing students with Malware analysis skills. Topics covered in this course are analysis of advanced contemporary malware using both static and dynamic analysis and using industry standard debugging tools, reverse engineer and machine code analysis how to identify vulnerabilities in executable code and construct effective defenses against malicious attacks, skills for reverse-engineering malicious software using a variety of freely available tools, how to identify and acquire investigative leads from host and network-based indicators associated with malicious code and applications. Hands-on is provided through a series of exercises, assignments, and projects.

Network Forensics (CSS425) **التحليل الجنائي للشبكات**

The course provides students with knowledge and practical skills essential for the acquisition, analysis and interpretation of evidence obtained from wired and wireless networked computers and storage systems. Topics covered in this course are the evidence harbored by network hardware devices, such as routers and switches, how to effectively retrieve valuable information from the Microsoft Windows Server operating systems and analyze the Active Directory for evidence pertaining to user policies, privileges, and violations, how to formulate and implement a strategy to acquire drive images as

evidence files over networked computers. Hands-on is provided through a series of exercises, assignments, and projects.

Cyber Defense (CSS426) **الدفاع السيبراني**

This course provides the fundamental principles of cyber defense. Topics covered in this course are security fundamental principles, vulnerability assessment, intrusion detection, cryptography protocols, network defense, and trust relationships; identification of reconnaissance operations, anomaly/intrusion detection, anomaly identification, identification of command and control operations, identification of data exfiltration activities, identifying malicious code based on signatures, behavior, and artifacts; networking security techniques and components (e.g., firewalls, IDS, etc.); cryptography (include PKI cryptography) and its uses in cybersecurity; malicious activity detection, system security architectures and concepts, defense in depth, and virtualization. Hands-on is provided through a series of exercises, assignments, and projects.

Selected Topics in Cyber Security – 1 (CSS428) **موضوعات مختاره في الأمن السيبراني - 1**

The course provides specific areas of interest and emerging topics within the field of Cyber Security. The course focuses on in-depth discussions and practical applications of selected topics, providing students with specialized knowledge and skills to tackle complex Cyber Security challenges. The specific topics covered may vary depending on the instructor and current trends of Cyber Security.

Selected Topics in Cyber Security – 2 (CSS429) **موضوعات مختاره في الأمن السيبراني - 2**

The course provides specific areas of interest and emerging topics within the field of Cyber Security. The course focuses on in-depth discussions and practical applications of selected topics, providing students with specialized knowledge and skills to tackle complex Cyber Security challenges. The specific topics covered may vary depending on the instructor and current trends of Cyber Security.

توصيف المقررات الدراسية علوم البيانات

Data Wrangling and Knowledge Discovery (ISD311) **معالجة البيانات واكتشاف المعرفة**

The course provides the principles, techniques, and tools involved in building and managing data wrangling and conducting knowledge discovery from large datasets. Topics covered in this course are data modeling, data extraction, transformation and loading (ETL), dimensional modeling, data integration, data quality, exploratory data analysis (EDA) for data wrangling, data wrangling with programming languages and tools, data wrangling for big data and data wrangling for machine learning. Hands-on experience is provided through a series of exercises, assignments, and projects.

Computational Linguistics (ISD416) **اللغويات الحاسوبية**

The course provides the students with the field of computational linguistics, which involves the development and application of computational models and techniques for understanding and processing human languages. Topics covered in this course are fundamentals of linguistics, natural language processing (NLP), text processing and tokenization, language modeling, syntax and parsing, machine translation, speech processing, sentiment analysis and opinion mining, ethics and bias in computational linguistics and future trends in computational linguistics. Hands-on experience is provided through a series of exercises, assignments, and projects.

Big Data Analytics (ISD411) **تحليلات البيانات الضخمة**

The course provides the students with the principles, tools, techniques that are used to analyze large and complex datasets for the purpose of extracting insights, patterns, and knowledge from massive amounts of structured and unstructured data. Topics covered in this course are introduction to Big data analytics, data acquisition and ingestion, Big data storage and management, data processing with Hadoop, Big data analytics with spark, data visualization for Big data, machine learning with Big data, real-time analytics and stream processing, Big data security and privacy, Big data ethics, real-world use cases of Big data analytics in different domains. Hands-on experience is provided through a series of exercises, assignments, and projects.

GIS and Spatial Data Mining (ISD413) **نظم المعلومات الجغرافية وتنقيب البيانات المكانية**

The course provides the concepts, techniques, and applications of Geographic Information Systems (GIS) and spatial data mining. Topics covered in this course are introduction to GIS and spatial data, spatial data visualization and cartography, spatial data acquisition and sources, spatial data preprocessing and cleaning, spatial data query and analysis, spatial data mining techniques, spatial data classification and prediction, spatial data privacy and ethics, and future trends in GIS and spatial data mining. Hands-on experience is provided through a series of exercises, assignments, and projects.

Data Visualization & Exploration (ISD421) **تصور البيانات وأستكشافها**

The course provides the students with the principles, techniques, and tools involved in visualizing data in a clear and compelling manner, as well as exploring data to uncover patterns, trends, and relationships. Topics covered in this course are fundamental concepts of data visualization, visual perception, data types, storytelling through data, data visualization tools and libraries, various visualization techniques, data visualization design, exploratory data analysis (EDA) and best practices

for creating visually appealing and informative visualizations. Hands-on experience is provided through a series of exercises, assignments, and projects.

Introduction to Block Chain & Distributed Ledgers (ISD415) **مقدمة في تقنية سلسلة الكتل والحسابات الموزّعة**

The course provides the students with a comprehensive understanding of the concepts, principles, and applications of blockchain technology and distributed ledgers. Topics covered in this course are fundamental components and mechanisms of blockchain systems and their potential to revolutionize various industry, the basic concepts of blockchain, including decentralization, cryptographic hashing, consensus mechanisms, and smart contracts and different types of blockchains, historical development of blockchain technology and its relationship to cryptocurrencies like Bitcoin and Ethereum and the advantages and limitations of blockchain systems, as well as the implications for privacy, security, and trust. Hands-on experience is provided through a series of exercises, assignments, and projects.

Data Compression Techniques (ISD425) **تقنيات ضغط البيانات**

This course provides the students with a comprehensive understanding of the principles, algorithms, and methods used to compress and decompress data efficiently. Topics covered in this course are introduction to data compression, information theory basics, run-length encoding (RLE), dictionary-based compression, statistical methods for compression, transform coding, image, audio and video compression techniques, lossless compression algorithms, performance evaluation of compression techniques, applications of data compression and future trends in data compression. Hands-on experience is provided through a series of exercises, assignments, and projects.

Theory and Practice of Data Analysis (ISD322) **نظرية وممارسة تحليل البيانات**

The course provides the students with a comprehensive understanding of the theoretical foundations and practical techniques used in data analysis also, extracting meaningful insights and making informed decisions based on data. Topics covered in this course are introduction to data analysis, data types and data preprocessing, descriptive statistics, probability and statistical distributions, hypothesis testing and statistical inference, regression analysis, data visualization techniques, dimensionality reduction, clustering techniques, machine learning for data analysis, time series analysis, ethics in data analysis and real-world data analysis case studies. Hands-on experience is provided through a series of exercises, assignments, and projects.

Distributed Data Analysis (ISD424) **تحليل البيانات الموزعة**

The course provides the students with the concepts, techniques, and tools used in analyzing large-scale distributed data. Topics covered in this course are introduction to distributed data analysis, distributed computing basics, distributed data storage, distributed data processing, data partitioning and shuffling, data serialization and compression, data pipelines and workflow management, distributed data analysis algorithms, handling streaming data, performance optimization and tuning, privacy and security in distributed data analysis and future trends in distributed data analysis. Hands-on experience is provided through a series of exercises, assignments, and projects.

Computational Statistics and Data Analysis (ISD321) **الإحصاء الحسابي وتحليل البيانات**

The course provides the students with the computational techniques and algorithms used in statistical analysis and data exploration for performing efficient and accurate statistical analysis. Topics covered in this course are introduction to computational statistics, data preprocessing and transformation, simulation techniques in statistics, computational linear and nonlinear regression, Bayesian data analysis, computational multivariate analysis, computational time series analysis, machine learning algorithms in computational statistics, Big data analytics with computational statistics, high-performance computing for statistical analysis and ethical considerations in computational statistics and data analysis. Hands-on experience is provided through a series of exercises, assignments, and projects.

Intrusion Detection and Prevention (ISD423) **أكتشاف والوقاية من التسلل والأختراقات**

The course provides the students with a comprehensive understanding of the principles, techniques, and technologies used to detect and prevent unauthorized access and malicious activities in computer systems and networks and ensuring the integrity, confidentiality, and availability of systems. Topics covered in this course are introduction to intrusion detection and prevention, network security fundamentals, intrusion detection system (IDS) basics, IDS technologies and tools, intrusion prevention system (IPS) techniques, firewalls and access control, machine learning for intrusion detection, signature-based intrusion detection, vulnerability assessment and penetration testing and legal and ethical aspects of intrusion detection. Hands-on experience is provided through a series of exercises, assignments, and projects.

Social Media Mining (ISD418) **التنقيب بوسائل التواصل الاجتماعي**

The course provides the students with a comprehensive understanding of the concepts, techniques, and applications used in the field of social media mining. Topics covered in this course are data collection, preprocessing, analysis, and visualization, algorithms and software related to mining of data available in social media, the state-of-the art developments in participatory Web techniques, social networks and analysis, information extraction, and Web mining, to study emerging problems with social media, and to learn innovatively applying multidisciplinary approaches to problem solving. Hands-on experience is provided through a series of exercises, assignments, and projects.

Applied Regression Methods (ISD412) **تطبيقات الطرق المرجعية.**

The course provides the students with the knowledge and skills to effectively model and analyze relationships between variables using regression techniques, in addition, understanding of regression analysis and its practical applications in various fields. Topics covered in this course are introduction to regression analysis, simple and multiple linear regression, model building and variable selection, interaction effects and nonlinear regression, categorical predictors in regression, model interpretation and inference, multicollinearity and VIF, model selection and regularization, time series regression, applications of regression analysis and implementing regression models on real datasets. Hands-on experience is provided through a series of exercises, assignments, and projects.

Applied Multivariate Analysis (ISD422) **التحليل التطبيقي متعدد المتغيرات.**

The course provides the students with the knowledge and skills to analyze and interpret complex datasets with multiple variables and introduces theories, techniques, and applications of multivariate statistical methods. Topics covered in this course are introduction to multivariate analysis, multivariate data visualization, multivariate descriptive statistics, multivariate normal distribution, principal component analysis (PCA), canonical correlation analysis (CCA), discriminant analysis, multivariate analysis of variance (MANOVA), structural equation modeling (SEM), applications of multivariate analysis, multivariate analysis with big data and software tools for multivariate analysis. Hands-on experience is provided through a series of exercises, assignments, and projects.

Optimization Algorithms (ISD414) **خوارزميات التحسين**

The course provides the students with both theoretical foundations and practical applications of optimization algorithms in diverse domains and explores various techniques used to optimize complex systems and solve optimization problems efficiently. Topics covered in this course are introduction to optimization, mathematical foundations for optimization, unconstrained optimization, constrained

optimization, linear programming, nonlinear programming, integer programming, combinatorial optimization, metaheuristic optimization algorithms, simulated annealing and tabu search, heuristic search techniques, optimization in machine learning, applications of optimization algorithms and optimization software and tools. Hands-on experience is provided through a series of exercises, assignments, and projects.

Selected topics in data science – 1(ISD426) **موضوعات مختارة في علوم البيانات - 1**

The course provides the students with specific areas of interest and emerging topics within the field of data science. The course focuses on in-depth discussions and practical applications of selected topics, providing students with specialized knowledge and skills to tackle complex data-related challenges. The specific topics covered may vary depending on the instructor and current trends in data science.

Selected topics in data science - 2 (ISD427) **موضوعات مختارة في علوم البيانات - 2**

This course provides the students with specific areas of interest and emerging topics within the field of data science. Building upon the foundational knowledge gained in the prerequisite courses, this course focuses on exploring advanced concepts, techniques, and applications in data science. The specific topics covered may vary depending on the instructor and current trends in data science.

توصيف المقررات الدراسية نظم المعلومات

Database Systems - 2 (ISI321) **نظم قواعد البيانات - 2**

The course provides the students with an in-depth understanding of the design and implementation of database systems and the administration features of any DBMS. Topics covered in this course are review of relational model, E-R and EER diagramming, normalization, SQL, relational algebra, query processing and optimization, transaction processing, concurrency control and recovery, database security and authorization, database architectures, distributed databases: architecture, distributed transaction processing, object oriented databases, data warehousing: heterogeneous component systems, data scrubbing, DW design and emerging database technologies and application such as XML and NoSQL. Hands-on experience is provided through a series of exercises, assignments, and projects.

Systems Analysis and Design -1 (ISI312) **تحليل وتصميم النظم - 1**

The course provides the students with the knowledge and skills necessary to analyze, design, and model information systems effectivelthey, in addition to providing students with the systematic approach to developing high-quality and efficient information systems to meet organizational needs. Topics covered in this course are the development life cycle in an organizational context identifying

and selecting system development project, information systems planning, requirement's elicitation process(traditional, contemporary) methods, requirements analysis and gathering, different information systems models, system analysis documentation and project management and overview of system design models.Hands-on experience is provided through a series of exercises, assignments, and projects.

Decision Support Systems (ISI421) **نظم دعم القرار**

The course provides the students with the knowledge and skills to understand, design, and use decision support systems effectively in various domains, such as business, healthcare, finance, and logistics. Topics covered in this course are Introduction to decision support systems, DSS components, DSS software and hardware, developing DSS, DSS models, types of DSS, executive information systems, data mining, artificial intelligence and expert systems, demonstrating DSS development approaches, utilizing DSS capacities to support different types of decision, Data-centered and Model-Centered DSS, DSS descriptive models, DSS predictive models and DSS optimization models. Hands-on experience is provided through a series of exercises, assignments, and projects.

Information Storage and Retrieval (ISI322) **تخزين واسترجاع البيانات**

The course provides the students with a comprehensive understanding of the principles, techniques, and technologies used to effectively store, organize, and retrieve information in various forms, such as text, and multimedia.Topics covered in this course are indexing techniques, consequential processing, the sorting of large files on disk, multilevel indexing, architecture of information storage and retrieval systems (IRS), models of IRS, user interface in IRS, text analysis, web search and cawling, ranking algorithms, Recent research in text retrieval and mining are emphasized and Text information retrieval systems. Hands-on experience is provided through a series of exercises, assignments, and projects.

Business Intelligence (ISI422) **ذكاء الأعمال**

The course provides the students with the knowledge and skills to effectively gather, analyze, and interpret business data to support decision-making and strategic planning within an organization.Topics covered in this course are overview of business intelligence, understanding the BI process and its role in organizational decision-making, principles of data visualization and information design, overview of analytical methods and techniques used in BI, descriptive, predictive, and prescriptive analytics, business analytics, decision support process in business, olap and data integration, business reporting and data visualization, business intelligence tools and platforms, and

case studies and examples of BI applications in different sectors. Hands-on experience is provided through a series of exercises, assignments, and projects.

Management Information Systems (ISI411) **نظم المعلومات الادارية**

The course provides students with fundamental knowledge of management information systems and their concepts including the use of information systems for management decision-making and the impact of information systems on management. Topics covered in this course are overview of management information systems, MIS model, MIS in a total CBIS environment, knowledge management, management information characteristics, functional applications of management information systems (MIS), integrated business processes, MIS role in enhancing decision-making and emerging technologies and trends in MIS. Hands-on experience is provided through a series of exercises, assignments, and projects.

Data Mining and Data Warehousing (ISI412) **التنقيب عن البيانات وتخزينها**

The course provides the students with the knowledge and skills to effectively analyze large datasets, extract valuable patterns and insights, and design efficient data storage solutions for supporting data-driven decision-making processes.Topics covered in this course are introduction to data warehousing and Data mining with (concept, aspects, and life cycle), data pre-processing techniques, architecture of data warehouse, data warehousing models, ETL processes (Extract, Transform, Load), building and maintaining data warehouses, OLAP operations, association rules, mining models (Classification, Clustering), Prediction models and Case-based and Rule-based reasoning. Hands-on experience is provided through a series of exercises, assignments, and projects.

Geographical Information Systems (ISI323) **نظم المعلومات الجغرافية**

The course provides the students with a comprehensive understanding of the principles, concepts, and applications of Geographical Information Systems (GIS) technology and equip students with the knowledge and skills to effectively work with spatial data, create maps, and apply GIS in various domains.Topics covers in this course are introduction to GIS, types of spatial data, spatial data representation, coordinate system, spatial referencing, spatial analysis and query, remote sensing, spatial data source, characteristics of satellite raster data and raster data modeling. Hands-on experience is provided through a series of exercises, assignments, and projects.

Enterprise Resource Planning (ISI413) **تخطيط موارد المؤسسة**

The course provides the students with a comprehensive understanding of Enterprise Resource Planning (ERP) concepts, principles, and implementation processes and equip students with the knowledge and skills necessary to effectively analyze business requirements, select appropriate ERP solutions, and successfully implement and manage ERP systems.Topics covered in this course are an overview for ERP, the ERP value for business, how to implement ERP model, different suits of ERP, ERP risk management, building ERP project cycle and ERP Project control.

E-Business and Digital Firm (ISI428) **الكترونيات الأعمال والشركات الرقمية**

The course provides a detailed overview of the concepts and processes used in doing business electronically and introducing principles, strategies, and technologies involved in conducting business in the digital era.Topics covered in this course are Introduction to digital firms and e-business, E-commerce for e-business, Infrastructure for e-business, Supply chain management, Collaborative Customer relationship management, Online retail, content, and services. Hands on Experience approach will enable students to gain valuable skills and practical knowledge in applying e-commerce technologies to real-life scenarios.

Systems Analysis and Design -2 (ISI429) **تحليل وتصميم النظم - 2**

The course provides the students with the knowledge and skills to apply advanced techniques and methodologies for analyzing, designing, and implementing effective information systems. Topics covered in this course are introduction to system design process, different design approaches and techniques, top-down design, bottom-up design, design patterns, system implementation and maintenance, system security threats, system testing, risk analysis and quality control. Hands-on is provided through exercises and projects to develop practical skills in requirements gathering, modeling, system design, and testing.

Software Projects Management (ISI324) **إدارة مشروعات البرمجيات**

The course provides the students with an introduction to the basic principles of managing a software development or maintenance project and introducing principles, methodologies, and best practices required to lead successful software projects from inception to completion.Topics covered in this course are building project plan, project authorization and stakeholders, project estimation and scheduling, plan optimization, risk management, quality management, project budgeting and accounting, project execution and changes management. Hands-on is provided through a series of exercises, assignments, and projects.

Health Care Information Systems (ISI416) **نظم المعلومات الصحية**

The course provides the students with the concepts and application of major information systems methodologies and approaches in the delivery of modern health care systems. Topics covered in this course are introduction to health care information systems, electronic health records, applications, standard strategic planning to health care information systems, Privacy and security of health care information systems, Types of health care information systemsand Performance standards and quality measures of health care information systems.

Operations Research (ISI325) **بحوث العمليات**

The course provides the students with the knowledge and skills of basic deterministic models and methods in operations research as well as stochastic models and use mathematical and analytical methods to optimize decision-making processes to improve the efficiency and effectiveness of operations in various domains.Topics covered in this course are linear programming, algebraic modelling and solutions, duality theory and sensitivity analysis, project scheduling, integer programming, dynamic programming, non-linear programming and game theory. Hands-on is provided through a series of exercises, assignments, and projects.

Expert Systems (ISI326) **نظم الخبرة**

The course provides the students with a comprehensive understanding of the principles, methodologies, and applications of expert systems and introducing the knowledge and skills to design, develop, and utilize expert systems effectively in various domains. Topics covered in this course are expert systems overview, knowledge acquisition, script representation, OAV-SN-Frames model, predicate logic, production rules, inference engine, inference network and uncertainty challenges and solutions. Hands-on experience is provided through a series of exercises, assignments, and projects.

Multimedia Database (ISI415) **قواعد بيانات الوسائط المتعددة**

The course provides the students with the characteristics and challenges of multimedia data, different types of multimedia data and their representations, and multimedia database design considerations. Topics covered in this course are introduction to multimedia databases, review of database approaches (relational, object oriented), multimedia data and data formats, querying multimedia, metadata creation and extraction of multimedia data, multimedia database architectures, distributed multimedia databases, approaches to indexing and querying text documents, multimedia retrieval (image, audio, video) and multimedia database applications. Hands-on experience is provided through a series of exercises, assignments, and projects.

Knowledge Base Systems (ISI427) **نظم المعرفة**

The course provides the students with the knowledge and skills to develop intelligent systems that can acquire, represent, and utilize knowledge to solve complex problems and support decision-making processes. Topics covered in this course are introduction to knowledge-based systems, knoweldge engineering, intelligent agents and case-based reasoning, knowledge based systems and search algorithms, ontological engineering, and semantic web, ID3 algorithms, fuzzy logic, probabilistic knowledge-based systems, machine learning and nueral networks. Hands-on is provided through a series of exercises, assignments, and projects.

Human-Computer Interface Development (ISI414) **تطوير واجهة المستخدم**

The course provides the students with a comprehensive understanding of the interaction between computer systems and the people who use them, introducing analysis and design techniques that can improve the quality of that interaction. Topics covered in this course are Introduction to human-computer interface, Task-centered system design, Methods for evaluation of interfaces with users, Byeond screen design, Graphical screen design, Design principles and usability heuristics and Principles of design standard. Hands-on is provided through a series of exercises, assignments, and projects.

Selected Topics in Information Systems -1 (ISI423) **موضوعات مختارة في نظم المعلومات -1**

The course provides specific areas of interest and emerging topics within the field of Information Systems. The course focuses on in-depth discussions and practical applications of selected topics, providing students with specialized knowledge and skills to tackle complex Information Systems challenges. The specific topics covered may vary depending on the instructor and current trends in Information Systems.

Selected Topics in Information Systems - 2 (ISI424) **موضوعات مختارة في نظم المعلومات -2**

The course provides specific areas of interest and emerging topics within the field of Information Systems. The course focuses on in-depth discussions and practical applications of selected topics, providing students with specialized knowledge and skills to tackle complex Information Systems challenges. The specific topics covered may vary depending on the instructor and current trends in Information Systems.

Introduction to 2D animation(DMT324) **مقدمة في الرسوم المتحركة ثنائية الابعاد**

The course provides the students with the basic knowledge of how to design and create motion graphics for multimedia and build projects appropriate both for internet applications and for film and video. Topics covered in this course are the principles of animation, creation of frame-by-frame animation, character design, creating and developing narratives with storyboarding, applying effects, and exporting animation to proper formats. The students will use a vector-based animation program and animation and visual effects compositing programs to create original work. An emphasis will be placed on planning projects and developing narratives through the use of storyboards.

Introduction to 3D animation (DMT411) **مقدمة في الرسوم المتحركة ثلاثية الابعاد**

The course provides the students with the knowledge and skills to construct 3D motion graphics for video, film, game, multimedia and internet applications. Topics covered in this course are model three-dimensional objects from idea generation, visual communication and character design, proficient development for life like character animations, through to the technical processes of modelling, texturing, animating, lighting, setting up cameras and rendering. Students will use an advanced CGI program to make original animations in a three-coordinate space.

User Experience Design (DMT326) **تصميم تجربة المستخدم**

The course provides the students with a deep understanding of user-centered design principles, usability testing, information architecture, and interaction design to create effective and user-friendly software apps. Topics covered in this course are how to conduct user research, analyze data, and translate insights into intuitive app interfaces. It will also cover the required skills to design wireframes, mockups, and interactive prototypes using industry-standard tools. By the end of the course, students will have the expertise to develop seamless and engaging app prototypes, ensuring they can deliver exceptional user experiences that meet the needs and expectations of modern software applications.

Video and Audio Technology (DMT322) **تكنولوجيا الصوت والصورة**

The course provides the students with a comprehensive understanding of the audio and video production industry and its post-secondary educational and career opportunities. Topics covered in this course are every aspect of several class and small group audio, video, and film style production projects with emphasis on live multi-camera video and film style production projects. Professional grade equipment and software will be used in the creation of student lead productions.Students will

gain job-specific training for entry level employment in audio, video, television, and motion picture careers.

Virtual Reality (DMT421) **الواقع الافتراضي**

The course provides the students with a comprehensive understanding of virtual reality (VR) techniques and tools, in addition to exploring the various emerging applications of VR in real-life activities. Topics covered are introduction to virtual environment; 3D geometric modeling and transformation; Free form deformation; Particle systems; Physical simulation; Human factors; VR hardware; VR software; VR applications. Throughout the course, students will work on hands-on projects, allowing them to apply their knowledge in creating interactive VR applications for various industries, such as gaming, education, training, and simulations.

Augmented Reality (DMT422) **الواقع المعزز**

The course provides the students with a comprehensive understanding of Augmented Reality(AR) principles, tools, and development methodologies. Topics covered in this course are how to blend digital content with the real world, creating seamless and interactive AR experiences marker-based and markerless tracking, object recognition, and spatial mapping. Students are engaged in hands-on projects to design and implement AR applications for various domains, including gaming, marketing, education, and industrial training.

Digital Signal Processing (DMT311) **معالجة الإشارات الرقمية**

The course provides the students with a comprehensive understanding of continuous time and discrete time signals and systems, linear time invariant systems. Topics covered in this course are fourier transform for continuous and discrete time signals, Sampling theorem, Laplace transform. Review of principles of discrete signals in time and frequency; Transform-domain representations of discrete time sequences; Fast Fourier transform; Structural representations of digital filters, Digital Filter design problems, Implementation aspect of digital signal processing algorithms, Introduction to filter banks and wavelets. Hands-on is provided through a series of exercises, assignments, and projects.

Game Development (DMT412) **تطوير الألعاب**

The course provides the students a comprehensive understanding of different AI techniques commonly used in gaming, such as pathfinding, decision-making, and behavior modeling. Also, the course allows the students to develop their critical thinking and problem-solving skills to design

challenging and engaging game scenarios using AI. Topics covered in this course are game production history and current practices for game development including software, environments and production process. Students will Analyze real-world case studies of successful games that effectively utilize AI for gameplay and storytelling. Students work individually and/or in groups to develop and create intelligent game prototypes, simulations and complete a video game throughout the semester.

Character Design for film and games (DMT325)

تصميم الشخصيات للأفلام والألعاب

The course provides the students with the fundamental principles of character design, including anatomy, gesture, expression, and personality development so the students will be able to conceptualize and visually communicate unique and believable characters that align with the storytelling requirements of both film and video games. Students will also gain insights into designing characters that resonate with specific target audiences and fit within the visual style of the project. Topics covered in this course are the entire character design pipeline, from initial sketches and iterations to detailed digital illustrations or 3D models. By the end of the course, students will possess the skills to craft exceptional characters that breathe life into narratives, adding depth and richness to the worlds of film and games.

Computer Graphics (DMT323) **الرسم بالحاسب**

The course provides the students with a comprehensive understanding of computer graphics concepts and an overview of graphics systems. Topics covered in this course are Line drawing algorithms; Circle drawing algorithms; Ellipse drawing algorithms; Area filling algorithms; Polygon filling algorithms; Line clipping algorithms; Polygon clipping algorithms; Two dimensional transformations; (translation, rotation, scaling , general transformations, composite transformations); Three dimensional object representation and Projections; Three dimensional modeling and transformations (translation, rotation, scaling, sheer, reflection, composite), Three dimensional Viewing and Camera Model. Visible surface detection algorithms; Reflection and illumination models. Hands-on is provided through a series of exercises, assignments, and projects.

Distributed Multimedia Applications (DMT423) **تطبيقات الوسائط المتعددةالموزعة**

The course provides the students a comprehensive understanding of multimedia application design in distributed computing environments. Topics covered in this course will include technologies like streaming media, real-time communication, and content synchronization, developing applications with various media types, addressing challenges such as network latency and scalability, and merging

trends like cloud-based multimedia and virtual reality. By the end of this course students are equipped to create sophisticated applications for entertainment, education, and more. Prerequisites: Prior experience in multimedia technologies, networking, and programming recommended.

Speech Processing (DMT425) **معالجة الكلام**

The course provides the students with a comprehensive understanding of speech processing as one of the main application areas of digital signal processing such as the new emerging technologies of voice over-IP, automated call centres, voice browsing and biometrics find commercial markets. Topics covered include introduction to speech production; general properties of speech signal; Time domain processing of speech; Frequency domain processing of speech; Linear prediction analysis; Cepstral analysis; Feature extraction for speech processing; Introduction to statistical speech recognition; Introduction to speech coding. Hands-on is provided through a series of exercises, assignments, and projects.

Visual Effects (DMT414) **مؤثرات بصرية**

The course provides the students with the art and technology of creating stunning visual effects for film, television, and other media. Topics covered in this course are the fundamental principles and techniques used in the visual effects industry including compositing, motion tracking, 3D animation, and digital simulations, green screen compositing, particle effects, creature animation, and more. Students will gain proficiency in industry-standard software tools, such as Adobe After Effects, Autodesk Maya, and Nuke, and anyother recent application work on practical projects to create realistic visual effects sequences. Additionally, students will explore the integration of visual effects with live-action footage and gain insights into the collaborative nature of visual effects production.

Video Processing (DMT416) **معالجة الفيديو**

The course provides the students with the fundamentals of digital video representation, compression, enhancement, and analysis where the students will learn how to manipulate and transform video data using image processing algorithms and tools. Topics covered include video filtering, motion estimation, object tracking, and video stabilization. Students will gain hands-on experience with industry-standard software and libraries for video processing. The course also covers advanced topics, such as video coding standards and video analytics for surveillance and computer vision applications. Through real-world projects and case studies, students will develop a deep understanding of video processing techniques and their role in modern multimedia and computer vision systems.

Selected Topics in Digital Media Technology -1 (DMT426)

موضوعات مختاره في تكنولوجيا الوسائط الرقمية-1

The course provides specific areas of interest and emerging topics within the field of Digital Media . The course focuses on in-depth discussions and practical applications of selected topics, providing students with specialized knowledge and skills to tackle complex Media Technology challenges. The specific topics covered may vary depending on the instructor and current trends in Digital Media Technology..

Selected Topics in Digital Media Technology -2 (DMT427)

موضوعات مختاره في تكنولوجيا الوسائط الرقمية-2

The course provides specific areas of interest and emerging topics within the field of Digital Media . The course focuses on in-depth discussions and practical applications of selected topics, providing students with specialized knowledge and skills to tackle complex Media Technology challenges. The specific topics covered may vary depending on the instructor and current trends in Digital Media Technology.

توصيف مشروعات التخرج

Project 1,2 (PR498,PR499) **مشروع 1,2**

The course will continue for two semesters. In the first semester, a group of students will select one of the projects proposed by the department, and analyze the underlying problem. In the second semester, the design and implementation of the project will be conducted.